MIL-G-83872A (USAF)
14 March 1977
SUPERSEDING
MIL-G-83872 (USAF)
12 January 1976

MILITARY SPECIFICATION

GYROSCOPE, DISPLACEMENT,

TYPE SBU-23/A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for one type of roll and pitch displacement gyroscope, one type of baseplate assembly designated MD-1 Gyro Replacement Baseplate Assembly, and one type of baseplate assembly designated PB-60 Autopilot Gyro Replacement Baseplate Assembly. When the roll and pitch displacement gyroscope is mated with the MD-1 Gyro Replacement Baseplate, it shall be designated a Type MD-1 (Replacement) Gyroscope. When the roll and pitch displacement gyroscope is mated with the PB-60 Autopilot Gyro Replacement Baseplate, it shall be designated a type PB-60 (Replacement) Gyroscope.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: (AFLC CASO/LODS, Federal Center, Battle Creek MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2 The Gyroscope shall also include internally the capability to perform the rate switching function integrally and simultaneously while performing the vertical displacement output function. This additional function shall be implementable through insertion of a contractor furnished short extension cable, connected between the gyroscope and the aircraft's gyro connector. When this extension is installed, it shall be possible to disconnect an aircraft's MC-1 rate switching gyro and still include rate switching function, in addition to vertical displacement gyro output functions. This method of connection flexibility is required to maintain gyroscope interchangeability with those aircraft equipped with MD-1 vertical gyro and MC-1 rate switching gyro. It is in addition to the interchangeability on aircraft equipped with PB-60 autopilot vertical displacement gyroscopes (Bendix Part Number 14161).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents - The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416	Plating, Cadmium (Electrodeposited)

MILITARY

MIL-P-116	Preservation-packaging, Methods of
MIL-C-4150	Case, Transit and Storage, Waterproof and Water-Vaporproof
MIL-C-5015	Connectors, Electric AN Type
MIL-B-5087	Bonding, Electrical, and Lightning Protection for Aerospace Systems
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment. Aircraft, General Specification for
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys

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MIL-P-19834	Plates, Identification, Metal Foil, Adhesive Backed
MIL-G-25591	Gyroscope, Rate Switching, Type MC-1
MIL-I-27193	Indicator, Attitude ARU-2B/A
MIL-I-27619	Indicator, Attitude Director
MIL-I-27623	Indicator, Attitude, Remote Standby, 2-inch
MIL-I-27709	Indicator, Attitude, ARU-14/A Remote
MIL-I-27710	Indicator, Attitude Remote
MIL-I-38442	Indicator, Attitude Director, ARU-24/A

STANDARDS

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specification, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics Measurement of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests Expotential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS 3101	Connector, Plug, Electric, Cable Connecting
MS 3106	Connector, Plug, Electric, Straight
MS 3108	Connectors, Plug, Electric, Solder Contact, 90 Degree
MS 14108	Fastener, Positive Self-Locking, Case Mounting, Electronic Equipment
MS 17322	Meter, Time Totalizing Miniature, Digital 115 Volt 400 Cycle
MS 27150	Indicator, Attitude, Remote Standby
MS 33678	Connector, Receptable, Electric, Integral Mounting

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification - The gyroscope assembly, electronics assembly, and the Type MD-1 and Type PB-60 baseplace assemblies furnished under this specification shall be a product, samples of which have been tested, passed the qualification tests specified herein, and have been listed on or approved for listing on the applicable qualified products list (See 4.2 and 6.5).

3.2 Components - The gyroscope shall consist of: (1) a gyroscope assembly in which gimbals are electrically erected and internally shockmounted. The gyroscope assembly shall be hermetically sealed. (2) an electronics assembly for the gyroscope. The electronics assembly may either be housed in the hermetically sealed chamber of the gyroscope assembly, or housed in an adjacent non-hermetically sealed chamber. (3) a base assembly (either for MD-1 type replacement or PB-[illegible] type replacement, as specified in the contract) which sh[illegible]e a non shockmounted, quick disconnect type. (4) an opti[illegible]cated Time Totalizing Meter. (5) a short extension cabl[illegible]e foot or less in length) for utilization of the rate sw[illegible]ing function described in paragraph 1.2.

3.2.1 Design - The gyroscope shall be designed in such a manner that pitch and roll autopilot outputs (sine, cosine, and versine functions), autopilot interlock, and internal roll erection cutout may be added and connector changed at field level by substituting a PB-60 Gyro base assembly for the MD-1 Gyro base assembly and with no changes to the roll and pitch displacement gyroscope.

3.2.2 Specifications and Standards - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3.1 Fungus-Proof Materials - Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment is unnecessary.

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3.3.2 Metals - Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.2.1 Dissimilar Metals - Dissimilar metals are defined by MIL-STD-889. Unless suitably protected against electrolytic corrosion by means of protective coating or hermetic sealing, intimate contact of dissimilar metals shall be avoided.

3.3.3 Nonmagnetic Materials - Nonmagnetic materials shall be used for all parts of the gyroscope, except where magnetic materials are essential.

3.3.4 Corrosive Fumes - The materials, as installed in the gyroscope and under the service conditions specified herein, shall not liberate deleterious fumes.

3.3.5 Protective Treatment - When materials are used in the construction of the gyroscope that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they should be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or peel with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and Construction - The gyroscope shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.4.1 Total Weight - The total weight of the gyroscope and MD-1 replacement base plate, excluding cables, shall be a minimum consistent with good design and shall not exceed 9.5 pounds. The total weight of the gyroscope and PB-60 replacement base plate shall be a minimum consistent with good design and shall not exceed 14 pounds.

3.4.2 Reliability

3.4.2.1 Operating Life (Longevity) - The gyroscope shall have an operating life, as defined per MIL-STD-781, of 18,000 hours minimum.

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3.4.2.2 Reliability in Mean Time Between Failures (MTBF) - The gyroscope shall be designed to provide a minimum acceptable (θ_1) MTBF of 2100 hours at a 90% confidence level as defined in MIL-STD-781.

3.4.2.3 Time Totalizing Meter - The gyroscope shall contain an MS-17322-6A, or equivalent, time totalizing meter. All operating hours referred to in this document shall be as recorded on the Time Totalizing Meter. The time totalizing meter shall be located in the gyroscope and shall be visible without unsealing the gyro.

3.4.3 Cabling and Connections

3.4.3.1 Cables and Connectors - The gyroscope shall provide for the use of cables and connectors in accordance with MIL-E-5400.

3.4.3.2 Interconnecting Cabling

3.4.3.2.1 MD-1 (Replacement) Gyroscope Assembly - The gyroscope shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088 and Figure 1, 2, or 3. The external wiring shall be unshielded except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements, and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the gyroscope except that Adapter Cable, Figure 3, shall be supplied to allow operation of the gyroscope with internal bank angle roll erection cut-out.

3.4.3.2.2 PB-60 (Replacement) Gyroscope Assembly - The PB-60 (replacement) gyroscope assembly shall be capable of satisfactory operation using external wiring in accordance with the requirements of MIL-W-5088 and Figure 7 (Ref J1 and J2). J3 of the PB-60 gyro replacement baseplate shall be attached to the gyroscope mounted upon this baseplate. The external wiring shall be unshielded except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assemhly of the cable to its plugs may be easily accomplished. The connectors shall be wired in accordance with Table 1A.

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3.4.3.3 External Connections - The MD-1 (replacement) gyroscope shall be so designed that it will operate as specified herein under the following conditions:

(a) When the external connections shown in Figure 1, 2, and/or 3 are made to the ARU/13A indicator conforming to MIL-I-27710, the ARU-14/A indicator conforming to MIL-I-27709, or the MS-27150 indicator conforming to MIL-I-27623.

(b) When the external connections shown on Figure 1, 2, and/or 3 are made by means of a banana-plug board by selecting the correct power, synchro, and warning flag pins for the ARU-2B/A indicator conforming to MIL-I-27193, the ARU-11/A indicator conforming to MIL-I-27619, or the ARU-24/A indicator conforming to MIL-I-38442.

3.4.4 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400.

3.4.5 Electromagnetic Interference - The gyroscope shall comply with the requirements of MIL-STD-461A, Notice 3, for Class A1 equipment for Test Methods CE03, CE04, CS01, CS02, CS06, RE02, RS02, and RS03.

3.4.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in MIL-E-5400.

3.4.7 Nameplates - Nameplate approval for equipment identification shall be in accordance with MIL-P-19834.

3.4.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

Temperature	Room Ambient (25 ± 5°C)
Altitude	Normal Ground Level
Vibration	None
Humidity	Room ambient up to 90% Relative Humidity

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Excitation MD-1 (Replacement) Gyro	115 ± 0.5 VAC, 3-Phase ± 5 Hz (Phase Sequence ABC from pins C to A to B, Figure 1)
PB-60 (Replacement) Gyro	115 ± 0.5 VAC, Single Phase, 400 ± 5 Hz

3.4.9 Service Conditions - The gyroscope shall operate satisfactorily under any of the environmental service conditions, or reasonable combination of these conditions as specified in MIL-E-5400 for Class 3 equipment, except as modified herein.

3.4.9.1 Vibration - The gyroscope shall be tested in accordance with the vibration error and vibration endurance tests of Section 4 herein, paragraphs 4.8.20 (and sub paragraphs) and 4.8.23.6 respectively.

3.4.10 Warm-Up Time - The time required for the equipment to warm up prior to operation shall be kept to a minimum and shall not exceed five minutes.

3.4.11 Primary Input Power Requirements

3.4.11.1 MD-1 (Replacement) Gyroscope Assembly - The power supply to the MD-1 (replacement) gyroscope assembly shall be from 3-phase delta, 400 Hz, 115 volts line to line. The phase sequence shall be ABC from pins C to A to B, Figure 1. However, the gyroscope assembly shall operate from single phase 115 volts. The gyroscope shall function with variations in voltage and frequency from 103 to 127 volts and 320 to 480 Hz respectively.

3.4.11.2 PB-60 (Replacement) Gyroscope Assembly - The power supply to the PB-60 (replacement) gyroscope assembly shall be single phase, 400 Hz, 115 volts. The assembly shall function with variations in voltage and frequency from 103 to 127 volts and 320 to 480 Hz respectively.

3.4.11.3 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

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3.4.12 Hermetic Sealing - The gyroscope assembly shall be hermetically sealed. The electronics assembly for the gyroscope may either be housed in the hermetically sealed chamber of the gyroscope assembly, or housed in an adjacent non-hermetically sealed chamber. The electronics associated with the autopilot outputs (sine, cosine, and versine functions), contained in the PB-60 gyro replacement baseplate, shall not require hermetic sealing.

3.4.12.1 Filling Medium - The filling medium of the hermetically sealed gyroscope assembly shall be 95 percent helium and 5 percent carbon dioxide. The gases used shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The absolute pressure of the filling medium in the case shall be 1 atmosphere.

3.4.12.2 Leak Rate - The leak rate of the gyroscope assembly hermetic seal shall not cause more than 10 percent contamination of the filling medium after 1000 hours at a pressure differential equivalent to a 100,000 - foot altitude.

3.4.12.3 Case - The gyroscope's hermetically sealed case shall be of such strength that 1 atmosphere of differential pressure will not cause binding of either the pitch or roll gimbals.

3.4.13 Wiring - The internal electrical wiring shall be insulated from its case.

3.4.14 Insulation Resistance - The insulation resistance shall be 50 megohms or more when 500 volts DC is applied between isolated electrical circuits and between electrical circuits and the chassis. Insulation resistance test shall not be made on circuits connected with roll or pitch erection circuits.

3.4.15 Dielectric Strength - The electrical insulation shall not break down with a test voltage of 500 volts rms at a frequency of approximately 60 Hz applied for 5 seconds between isolated electrical circuits and between isolated electrical circuits and case. Dielectric tests shall not be made on circuits connected with roll or pitch erection circuits.

3.5 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service

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conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.5.1 Operation - The gyroscope shall provide gyroscopic stabilized roll and pitch reference data for aircraft utilization. The data shall be standard synchro type signals specified herein.

3.5.2 Performance at Environmental Extremes - Performance deviations at environmental extremes shall be permitted as specified in the detailed requirements.

3.6 Detailed Requirements

3.6.1 Displacement Gyroscope Assembly - The gyroscope assembly shall develop electrical signals representative of aircraft attitude through 360° roll and ± 82° of pitch. The gyroscope assembly shall consist of an electrically driven rotor. The gyro gimbals shall be electrically erected to the local gravity vertical. The gyroscope assembly gimbaling shall be internally vibration isolated and hermetically sealed to withstand the normal shocks, vibrations, and such other conditions incident to service, shipping, storage, and installation.

3.6.1.1 Dimensions

3.6.1.1.1 MD-1 (Replacement) Gyroscope Assembly - The dimensions of the gyroscope and MD-1 gyro replacement baseplate shall be in accordance with Figure 5.

3.6.1.1.2 PB-60 (Replacement) Gyroscope Assembly - The dimensions of the gyroscope shall conform to Figure 5 (excluding the baseplate). The dimensions of the PB-60 gyro replacement baseplate assembly shall be in accordance with Figure 8.

3.6.1.2 Gimbal Freedom - Unlimited freedom shall be provided about the roll axis and approximately ± 82° of freedom about the pitch axis. Pitch stops shall be incorporated which produce a roll precession when the pitch freedom is exceeded. The pitch stops shall be so designated that no parts of the instrument will be damaged by a loop at the rate of 90° per second and less than ± 3° error will result from loops of 10 ± 1° per second. Errors induced by roll maneuvers at 400° per second shall not exceed 1°.

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3.6.1.3 Drift - The gyroscope shall be so designed that the maximum free drift about each axis, noted after compensation for the earth's rotation, shall not be greater than 0.25° per minute at room temperature, with the gyroscope spin axis tilted 20° from vertical or at the vertical position. It shall not be greater than 0.5° per minute over the temperature range of -54°C to + 85°C.

3.6.1.4 Erection - The erection mechanism shall incorporate separate systems for pitch and roll. Vertical reference shall be established by means of two gravity sensitive devices which control a torque motor for each gyroscope axis. The erection system shall provide both fast and normal erection and shall erect the gyroscope from any attitude to the vertical. When the vertical is displaced in one axis, it shall not deviate more than 1° in the other axis while it is erecting.

3.6.1.4.1 Initial Erection - The gyroscope shall have fast initial erection, such that it erects to a position within 1° of vertical in pitch and roll within 60 seconds at ± 25°C. It shall erect to this position in less than 90 seconds at the temperature extremes of -54°C to + 85°C or power variation. At low temperature, low voltage and low frequency power, the initial erection time shall be 120 seconds maximum. The gyroscope shall be capable of meeting these initial erection requirements for the conditions in which the rotor is at standstill and the spin axis is anywhere within the limits of ± 85° bank and ± 82° pitch from the vertical at the time power is applied.

3.6.1.4.2 Final Erection - The gyroscope spin axis shall erect to within 0.25° of vertical in pitch and roll within 2 minutes after the initial erection period.

3.6.1.4.3 Erection Rate

3.6.1.4.3.1 Normal Erection Rate - The rate of normal erection about each axis shall be 2° ± .5° per minute at room ambient conditions, with a proportional decrease to 0° per minute starting at ± .5° vertical error from true vertical as shown by the limits of Figure 4. Over the temperature range of -54°C to + 85°C, the rate shall be 2° ± 1° per minute. The gyroscope shall be capable of meeting these erection rate requirements throughout all possible combinations of roll and pitch attitudes of the gyroscope.

3.6.1.4.3.2 Fast Erection Rate

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3.6.1.4.3.2.1 MD-1 (Replacement) Gyroscope Assembly - With the gyroscope rotor at full speed, the gyroscope shall have fast erection capability of not less than 15° per minute at any temperature between -54°C and + 85°C when pins N and C of the connector are shorted. In addition, the circuit between pins J and T shall be interrupted as long as pins N and C are shorted.

3.6.1.4.3.2.2 PB-60 (Replacement) Gyroscope Assembly - With the gyroscope rotor at full speed, the gyroscope shall have fast erection capability of not less than 15° per minute at any temperature between -54°C and + 85°C, when pins J1-h and J2-A of the connectors are shorted. In addition, the circuit between pins J1-M and J1-P shall be interrupted as long as pins J1-h and J2-A are shorted.

3.6.1.4.4 Erection Cutout

3.6.1.4.4.1 MD-1 (Replacement) Gyroscope Assembly - The design shall provide for cutout of pitch and roll erection external to the gyroscope, see Figures 1 and 2. The design shall also provide for internal roll erection cutout of the gyroscope whenever pins W and X are shorted together, pin L receives no input, and a bank angle of 7° ± 1° is exceeded (see Figure 3 for a description of the adaptor cable capable of accomplishing this). The automatic roll erection cutoff shall re-cycle when the bank angle is less than 4° ± 1° or after roll erection has been cutoff for more than 5 minutes. Whenever the bank angle becomes less than 4° ± 1°, the 5 minute enable period shall be reset. When the gyro receives external roll and pitch erection cutout, the current drawn from pin B to each pin (L and K) shall not exceed 250 ma during erection cutout.

NOTE: Manual Fast Erection should override Internal Bank Angle Cutout, but not External Erection Cutout (MC-1 rate switching function).

3.6.1.4.4.2 PB-60 (Replacement) Gyroscope Assembly - The PB-60 assembly design shall provide for automatic roll erection cutout whenever a bank angle of 7° ± 1° is exceeded. The automatic roll erection cutoff shall re-cycle when the bank angle is less than 4° ± 1° or after roll erection has been cut off for more than 5 minutes. Whenever the bank angle becomes less than 4° ± 1°, the 5 minute enable period shall be reset. The design shall provide for the optional external roll and pitch erection cutout of the gyroscope. See Table 1A.

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3.6.1.4.5 Manual Torquing - The gyroscope assembly shall be capable of torquing the spin axis to simulate a climb attitude and a left back attitude. See Table 1 for the MD-1 replacement gyroscope assembly, and Table 1A for the PB-60 replacement gyroscope assembly. This external torquing shall be obtained by connecting pin G to pin C of the MD-1 replacement gyroscope assembly, and pin J2-J to pin J2-A of the PB-60 replacement gyroscope assembly. Erection cutout on either axis and fast erection capability shall be available during manual torquing. During manual torquing the gyro top shall precess to the starboard and forward.

3.6.1.4.5.1 Torquing Rate - The gyroscope, when switched to the manual torquing mode, shall torque the spin axis at a minimum rate of 1° per minute.

3.6.1.5 Power Interruption - Means shall be provided in the gyroscope to prevent it from re-cycling through initial erection when power has been removed. After 5 minutes of normal erection, a power interruption shall not cause the gyroscope to develop errors in verticality exceeding $\pm$ 1° for a minimum of 40 seconds after interruption of power. Gyroscope gimbal freedom shall be not less than $\pm$ 82° in pitch and $\pm$ 80° in roll during this period. When power is restored, 40 seconds after interruption, the gyroscope shall not re-cycle.

3.6.1.6 Power Failure Warning - A power failure warning system shall be incorporated in the gyroscope which will close a set of contacts at the completion of the start cycle and open them when a power failure occurs, or manual fast erection is enabled.

3.6.1.7 Synchro Transmitter Output - Pickoffs of the synchro transmitter type shall be provided for both roll and pitch axis. The outputs shall be continuous for 360° roll and $\pm$ 82° pitch with the synchros following these gimbal movements with a one-to-one ratio of angular displacement. The electrical signals shall produce an unloaded output of 11.8 volts $\pm$ 0.2 volt line-to-line, 5.5° maximum phase shift. Each synchro shall be capable of operating up to five control transformers, each with an impedance of 222 + j 470 ohms (Zso). When loaded with five control transformers the synchro output shall not be less than 10.4 volts RMS line-to-line, phase shift not greater than 9°. Synchro nulls, with up to five loads shall not be greater than 30 mv. At temperature extremes, the no-load output of the synchros shall be 11.8 volts $\pm$ 0.3 volt, phase shift less than

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8°. When loaded with five control transformers, at temperature extremes, the line-to-line output of the synchro shall be greater than 10.2 volts with a mixiumum phase shift of 11.5°. The input voltage shall be 115 volts ± 0.25 volt, 400 ± 1 Hz for the synchro requirements.

NOTE: Concerning synchro output voltages applicable to the PB-60 replacement gyroscope assembly, the PB-60 assembly contains an internal load on both roll and pitch synchros. This constitutes one CT (control transformer) loading; therefore, voltage and phase shift tolerances can be compensated accordingly.

3.6.1.7.1 Synchro Static Accuracy. The static accuracy of the pitch and roll outputs, when the synchros are loaded with (5) control transformers specified in paragraph 3.6.1.7 shall be as follows:

Roll

Range	0° to ±10°	±10° to ±30°	±30° to ±180°
Accuracy	0.25°	0.35°	0.5°

Pitch

Range	0° to ±10°	±10° to ±30°	±30° to ±82°
Accuracy	0.25°	0.35°	0.5°

3.6.1.7.2 Synchro Designations

3.6.1.7.2.1 MD-1 (Replacement) Gyroscope Assembly - The synchro transmitter leads shall be defined with reference to the external electrical receptacle pins and ARINC designations as follows:

ROLL SYNCHRO		PITCH SYNCHRO	
Receptacle Pin	ARINC Designations	Receptacle Pin	ARINC Designations
D	X	R	X
E	Y	H	Y
F	Z	P	Z
M	H	U	H
S	C	V	C

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With the gyroscope case level and the gyroscope spin axis vertical, the synchro outputs shall be at the following ARINC designations:

	PITCH	ROLL
Index Reference Position	180°	180°
Positive Rotation	Climb	Right Bank

3.6.1.7.2.2 PB-60 (Replacement) Gyroscope Assembly - The synchro transmitter leads shall be defined with reference to the external electrical receptacle pins and ARINC designations as follows:

ROLL SYNCHRO		PITCH SYNCHRO	
J1	ARINC	J1	ARINC
E	X	H	X
D	Y	J	Y
F	Z	K	Z

With the gyroscope case level and the gyroscope spin axis vertical, the synchro outputs shall be at the following ARINC designations:

	PITCH	ROLL
Index Reference Position	0°	0°
Positive Rotation	Climb	Right Bank

3.6.1.7.3 Autopilot Outputs (PB-60 Replacement Gyroscope Assembly) - A control transolver shall be connected to each output synchro of the gyroscope assembly. The output of each transolver shall be two windings 90° apart with one winding proportional to the sine function of the attitude angle and the other to the cosine function. The rotor-stator of each transolver shall be locked together and oriented to supply proper bank and pitch angle signals for autopilot use. See Table 1A for required outputs.

3.6.1.7.4 Transolver Sine Outputs (PB-60 Replacement Gyroscope Assemble) - The sine outputs of the pitch and roll transolvers shall be oriented to 0° attitude. The signal gradient at 0° attitude shall be 200 mv per degree ± 1%. The phase of the output shall be as follows and shall not exceed ± 7° phase shift.

Right Bank	In Phase
Climb	In Phase

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3.6.1.7.5 Transolver Cosine Output (PB-60 Replacement Gyroscope Assembly) - The cosine output of roll shall be combined with the cosine output of pitch to produce a versine signal. The versine function shall increase in magnitude and climb phase as the bank changes in either direction by a value of 6000 (1 - Cos Ø) mv/degree.

3.6.1.8 Electrical Connector - The electrical connector provided on the gyroscope shall be a MS-33678-24-28P or equivalent and shall be located as shown in Figure 5 and shall be connected as shown in Table 1. All electrical circuits shall be isolated from the case.

3.6.1.9 External Test Points

3.6.1.9.1 MD-1 (Replacement) Gyroscope Assembly - The gyroscope shall be provided with the following minimum external test points:

C - N	(Manual Fast Erection)
K - B	(Pitch Erection Cutout)
L - B	(Roll Erection Cutout)
G - C	(External Torquing)
W - X	(Internal Roll Erection Cutout)

3.6.1.9.2 PB-60 (Replacement) Gyroscope Assembly - The PB-60 replacement gyroscope assembly shall be provided with the following minimum external test points:

J1-h - J2-A	(Manual Fast Erection)
J2-F - J2-A	(Pitch Erection Cutout)
J2-K - J2-A	(Optional External Roll Erection Cutout)
J2-J - J2-A	(External Torquing)

3.6.2 Dynamic Brake - The gyro shall be equipped for dynamic braking of its spin motor. The brake shall have the characteristics listed below. Figure 10 is a circuit example to fulfill these requirements.

a. Braking shall be accomplished through application of direct current electrical power (24 to 28 volts) between pins Q(+) and Z(-) of the gyro connector.

b. The brake circuit shall only activate when AC operating power is interrupted.

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c. It shall be capable of slowing the gyro spin motor from full synchronous speed to a speed approximately 5% or less of that value within 50 seconds after application.

d. It shall permit the gyro's fast erect circuit to automatically reactivate upon reapplication of AC operating power, provided the brake circuit has been permitted to function for at least 40 seconds before the reapplication of AC power.

e. The brake circuit shall be configured to automatically reset within 60 seconds of operation so that load on the direct current power source will be reduced to less than 0.1 ma.

f. The brake circuit shall have no effect on normal gyro function other than to provide the spin motor braking action when activated.

g. Circuit provisions shall insure that direct current is blocked (as by capacitors) from being introduced onto electrolytic switch circuits within the gyro.

h. The direct current braking circuit shall be applied to only one winding of the spin motor to accomplish the braking function.

3.7 Base Assembly

3.7.1 MD-1 (Replacement) Gyroscope Assembly - A base shall be provided for the mounting of gyroscope and electronic assemblies. It shall be designed for quick and easy attachment of the assemblies. It shall provide for easy electrical connection to the gyroscope. A self-locking, quick disconnect fastener, conforming to MS 14108 or equivalent shall be provided on the base. Means shall be provided to insure positive directional position location of the gyroscope on its base assembly. For outline dimensions, refer to Figure 6.

3.7.2 PB-60 (Replacement) Gyroscope Assembly - A base shall be provided for mounting the Universal gyroscope to form the PB-60 replacement gyroscope assembly. It shall be designed for quick and easy attachment of the gyroscope assembly. A self-locking, quick disconnect fastener, conforming to MS 14108 or equivalent shall be provided on the base. It shall provide for easy electrical interconnect between the gyroscope and base assembly and for easy electrical connection of the base assembly to existing aircraft wiring. For schematic of base

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plate refer to Figure 7. For outline dimensions refer to Figure 8. For installation clearance outline refer to Figure 9. Means shall be provided to insure positive directional position location of the gyroscope on its base assembly.

3.7.2.1 Total Weight - The total weight of the base plate shall be minimum consistant with good design and shall not exceed 5.5 pounds.

3.7.2.2 Design - The base plate shall be designed in such a manner that pitch and roll autopilot outputs (sine and cosine output) and automatic roll erection cutout are added to the basic gyro by the mounting of the basic Universal Gyroscope on the base plate assembly.

3.7.2.3 Interconnection Cabling - The base plate assembly shall be capable of satisfactory operation with the basic gyroscope using external wiring in accordance with the applicable requirements of MIL-W-5088 and Figure 7. The connectors shall be wired in accordance with Table 1A. Table 1B lists the pins which are interconnected in the base plate assembly.

3.7.2.4 Dimensions - The dimensions of the replacement PB-60 base assembly shall be in accordance with Figure 8.

3.7.2.5 Electrical Connectors - The electrical connectors provided on the PB-60 (replacement) base-plate assembly shall be PT06A-18-32P, PT06A-12-10P, and MS 3106A-24-28S-C or equivalent and shall be connected as shown in Figure 8. Connections shall be made as shown in Tables 1A and 1B.

3.8 Mounting Errors - When the gyro is mounted on its base plate, the internal center of gravity shock mounts and base plate shall not contribute more than + .75° dynamic verticality error, or ± .5° static error to the output of the gyro assembly.

3.9 Soldering - Soldering shall be accomplished in accordance with MIL-STD-454.

3.10 Screw Threads - Unless otherwise specified, the threads of all machine screws shall conform with MIL-S-7742.

3.11 Finishes and Protective Coatings

3.11.1 Aluminum Alloy Parts - Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

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3.11.1.1 Small holes and case inserts need not be anodized.

3.11.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.11.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.11.1.4 Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.11.1.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.11.1.6 Parts enclosed in a hermetically sealed container need not be anodized or plated.

3.11.1.7 When the part is plated with tin over a copper flash, the part need not be anodized.

3.11.1.8 When necessary for electrical bonding, parts need not be anodized.

3.11.2 Steel Parts - Where practicable, steel parts shall be cadmium plated in accordance with QQ-P-416, Type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.12 Workmanship - The gyroscope, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.12.1 Dimensions and Tolerances - Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the gyroscope, they shall be held or limited accordingly.

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3.12.2 Screw Assemblies - Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw, bolt, or threads.

3.12.3 Riveting - Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.12.4 Gears - Gear assemblies shall be properly aligned and meshed and shall operate without interference, tight spots, loose spots, or other irregularities. Where required for accuracy adjustment, gear assemblies shall be free from backlash.

3.12.5 Cleaning - The gyroscope shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips or other foreign material removed after assembly. Burrs and sharp edges, as well as resin flash which might crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests, as appropriate, to determine compliance with all applicable requirements.

a. Qualification Tests (4.2)
b. Initial Production Tests (4.3)
c. Acceptance Tests (4.4)
d. Reliability Qualification Test (4.8.25)

NOTE: Each Gyroscope shall be capable of passing all tests specified herein when attached/connected to the appropriate baseplate (MD-1 replacement or PB-60 replacement).

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4.2 Qualification Tests for QPL - Qualification for QPL shall consist of successful completion of tests in 4.8.1 through 4.8.24 plus 4.8.26; and, reliability qualification, either through flight test as described in 4.2.1 or through test as specified in 4.8.25.

4.2.1 Flight Test or Reliability Qualification Test - Fifteen gyroscopes with applicable baseplate assemblies shall be submitted to and pass the Reliability Qualification Test of 4.8.25, or the Flight Test described herein. The base-plates and 5 of the PB-60 autopilot gyro replacement baseplate assemblies so that the gyroscope assemblies submitted to either of these tests shall be comprised of 10 MD-1 (Replacement) Gyroscopes assemblies and 5 PB-60 (Replacement) Gyroscope assemblies. The flight test shall consist of these 15 gyroscope assemblies in service or military aircraft, performing routine worldwide missions. The gyroscope assemblies collectively shall demonstrate a mean time between failures (MTBF), in total operating time, of at least 2100 hours (unbiased); and, an MTBF in flying time of at least 1400 hours (unbiased). This must be accomplished for a total accumulated average operating time per gyro of not less than 2100 hours and a total accumulated average flying time per gyro of not less than 1400 hours. The Government must participate in the data collection and data reduction throughout the span of this test. For this reason, permission to qualify under the flight test requirement, rather than the reliability qualification test of 4.8.25, must be coordinated with the appropriate Government agencies who would be involved in support of such a test effort, and will depend upon approval and agreement of those agencies to participate. Any contractor desiring to submit to flight test must furnish the above 15 gyroscope and baseplate configurations at his own expense; and, he will be required to repair gyros throughout the period of the test and return them for continuation of test, all at his own expense. In addition, he will be required to submit to the Government, complete teardown and repair reports (TDRs), at contractor's expense, on each gyro assembly returned to him throughout the span of the test. Qualification of the flight test performance status shall not be granted until fulfillment of these requirements. Any contractor not having completed such a flight test successfully, must demonstrate flight and system suitability and safety (Paragraph 4.8.24), and perform all tests required herein in order to qualify his product.

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4.2.2 Test Samples - Five gyroscopes manufactured in accordance with the requirements of this specification, submitted at no cost to the Government, shall be subjected to the tests in Paragraph 4.8, except that 15 gyroscopes shall be used in the reliability test of Paragraph 4.8.25 and sub paragraphs, or a flight test. These tests shall be conducted at an activity designated by the agency responsible for granting qualification approval. All tests must be satisfactorily completed prior to qualification approval being granted (See 6.5). Production of gyros prior to approval shall be at the manufacturer's risk. Gyroscopes submitted to these tests shall not be shipped as production equipment.

4.2.3 Test Report - After the contractor completes the qualification tests, he shall prepare a test report in accordance with MIL-STD-831 and furnish three complete copies of the report to the activity designated by the agency responsible for granting qualification approval.

4.3 Initial Production Tests - When required by contract, 5 of the first 30 production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed the individual tests. The qualification samples shall not be selected for this test.

4.3.1 Scope of Tests - Initial production tests shall consist of the tests listed in Paragraph 4.8.

4.3.2 Initial Production Test Sample - Disposition shall be at the discretion of the procuring activity.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

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a. Individual Tests (4.4.1)

b. Production Reliability Assurance Test (4.4.2)

4.4.1 Individual Tests - Each gyroscope assembly (MD-1 replacement or PB-60 replacement) submitted for acceptance shall have been subjected to and passed the following applicable Individual Tests:

Test/Examination	Applicable Paragraph
a. Examination of Product	4.8.1
b. Initial Erection	4.8.2
c. Final Erection	4.8.3
d. Normal Erection Rate	4.8.4
e. Manual Fast Erection	4.8.5
f. Erection Cutout	4.8.6
g. External Torquing	4.8.7
h. Power Interruption	4.8.8
i. Power Failure Warning	4.8.9
j. Gimbal Freedom	4.8.10
k. Drift	4.8.11
l. Power Consumption	4.8.12
m. Synchro Requirements	4.8.13
n. Insulation Resistance	4.8.14
o. Leakage Tests	4.8.15
p. Dynamic Brake	4.8.26

4.4.2 Production Reliability Assurance Test.

4.4.2.1 Infant Mortality Burn-In Period - Each gyroscope submitted for acceptance shall be operated for 24 hours on a scorsby-type mechanism having a 15° movement at 5 to 7 oscillations per minute under test level "F" of MIL-STD-781, except that the vibration shall be 1 G. The 24 hour burn-in test shall consist of three 8-hour cycles (3 hours off and 5 hours on) with the following temperature cycling: The time to stabilize at lower temperature shall be three hours, and four hours to stabilize at high temperature and one hour operation at higher temperature. In the event of failure of any component or an out-of-tolerance condition of the gyroscope, the unit shall be unsealed and opened. The failed component shall be replaced or the out-of-tolerance condition shall be corrected and recorded as a nonrelevant failure. An investigation of the nonrelevant failure or out-of-tolerance condition that shows poor design or manufacture practices, shall be recorded and reported to the procuring activity along with corrective action.

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The unit shall then be resealed and continued on test for the time necessary only to complete a 24 hour burn-in period. At the conclusion of this test, the gyroscope shall be returned to the production lot and subject to selection at random for the acceptance sampling test, if applicable.

4.4.2.2 Sampling Test

4.4.2.2.1 Sampling Plan A - Unless otherwise specified (see 6.4c), 15 gyroscopes shall be selected at random from the first 50 items on the contract or order and tested as specified herein. The accept-reject criteria shall be in accordance with Test Plan V of MIL-STD-781, and shall demonstrate a minimum acceptable MTBF (θ_1) of 2100 hours. Test level F of MIL-STD-781 shall be used, except the vibration shall be 1G and the temperature extremes shall be +35°C and -10°C. The time to stabilize at lower temperature shall be 2 hours, and 3 hours to stabilize at higher temperature and 7 hours operating at higher temperature for a total of 12 hours for each cycle. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, its parts, and/or its workmanship.

4.5 Equipment Failure - Should a failure occur during tests described in Paragraph 4.3, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case, or a design defect.

c. Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

d. Where practical, include a test in the individual tests to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.6 Test Procedures - The procedures used for conducting qualification tests, acceptance tests, and reliability tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity, or the Government Inspector, to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract.

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4.7 Test Conditions

4.7.1 Atmospheric Conditions - Unless overwise specified, all the tests required by this specification shall be made in accordance with the requirements of 3.4.8.

4.7.2 Gyroscope - The gyroscope shall be installed on the Quick-Disconnect Mounting Base for all tests. All references to the gyroscope during the tests shall include the gyroscope Assembly, Electronics Assembly and the Mounting Base Assembly (MD-1 or PB-60 as applicable).

4.7.3 Attitude - Unless otherwise specified the gyroscope shall be mounted and mechanically secured to a platform which is level within 0.1° and is capable of 360° rotation in roll and pitch. The pitch and roll axis of the gyroscope shall be aligned to the pitch and roll axis of the platform.

4.7.4 Excitation

4.7.4.1 MD-1 (Replacement) Gyroscope Assembly - Excitation from 115 ± 0.5 VAC, 3 - Phase, 400 ± 5 HZ voltage input. The phase sequence shall be A to B to C. For the synchro test, excitation shall be 115 ± 0.25 VAC, 3 - phase, 400 ± 1 Hz. When operating the gyroscope on single phase operation (Figure 2) connect Pin Y to Pin A for proper failure detection.

4.7.4.2 PB-60 (Replacement) Gyroscope Assembly - Excitation from 115 ± 0.5 VAC, 400 ± 5 Hz, single phase. For the synchro test, excitation shall be 115 ± 0.25 VAC, 400 ± 1 Hz, single phase.

4.7.5 Test Equipment and Connections - Special Test Equipment and Connectors thereto shall be as specified. Whenever practical, the test equipment used in performing the tests shall not affect the performance of the gyroscope under test. For the MD-1 (replacement) gyroscope assembly, unless otherwise specified, the gyroscope shall be connected to an indicator having an input impedance, as near as practicable, equivalent to the ARU-14/A indicator in accordance with MIL-I-27709.

4.7.5.1 Normal Operation - Unless otherwise specified, normal operation shall be with power applied as in 4.7.4, with terminations as indicated in Table I for the MD-1 (replacement) gyroscope assembly, or Table 1A for the PB-60 (replacement) gyroscope assembly.

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4.7.6 Warm-Up Time - Unless otherwise specified, a five minute warm-up shall be allowed before the conduction of any tests. The warm-up test time of five minutes shall not waive the initial and final erection time for operational requirements.

4.8 Test Methods

4.8.1 Examination of Product - The gyroscope assembly (MD-1 replacement or PB-60 replacement) shall be inspected to determine compliance with this specification with regard to material, workmanship and marking. All tests shall be performed upon the entire applicable assembly.

4.8.2 Initial Erection - The gyroscope (including mounting base assembly) shall be mounted on a surface that is level within 6 minutes of arc. The gyroscope shall be displaced to any position in roll and to any position within ± 82° in pitch. With the gyroscope connected for normal operation, power shall be applied and the gyroscope allowed to erect. The gyroscope spin axis shall erect to ± 1° of vertical within 60 seconds at 25°C. At high temperature or power variation, the gyroscope spin axis shall erect to ± 1° of vertical within 90 seconds after application of power. At low temperature and low voltage and low frequency, the initial erection time shall be 120 seconds maximum. The end of the initial erection period is marked by the closing of the circuit between pins J and T for the MD-1 replacement gyroscope assembly, or between pins J1-M and J1-P for the PB-60 replacement gyroscope assembly.

4.8.3 Final Erection - The gyroscope shall be allowed to erect for two minutes after the initial erection period. At the end of the final erection period the spin axis shall be within ± 1/4° of vertical. (This test shall be performed without the benefit of a Scorsby table). The gyroscope shall then be rapidly rotated 180° in azimuth. The pitch and roll synchro out-puts shall not change by more than 1/2°.

4.8.4 External Torquing

4.8.4.1 MD-1 Replacement Gyroscope Assembly - The gyroscope assembly may be mounted on a Scorsby Table as in 4.8.5 and the gyroscope operated normally. Disconnect pins W and X. Connect pin G to pin C. The spin axis shall precess in a simulated climb and left bank attitude at a minimum rate of 1.0° per minute. Pin L shall be connected to pin B and the spin axis

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shall cease to precess in simulated left bank. Pin K shall be connected to pin B and the spin axis shall cease to precess in simulated climb. The connection from pins L and K, to pin B shall be removed and pin N shall be connected to pin C. The spin axis shall precess in simulated climb and left bank as above but at a rate of not less than 15° per minute. Connecting pins L and K to pin B shall cause precession to cease as specified above. During this test, the synchro outputs from the gyroscope shall give a direct indication of gyroscope position at all times.

4.8.4.2 PB-60 Replacement Gyroscope Assembly - The gyroscope assembly may be mounted on a Scorsby Table as in 4.8.5 and the gyroscope operated normally. Disconnect pins J3-X and J3-W. Connect pin J2-J to J2-A. The spin axis shall precess in a simulated climb and left bank attitude, at a minimum rate of 1.0° per minute. Cant the gyroscope assembly in a roll attitude; the spin axis shall cease to precess in a simulated left bank when the gyroscope assembly case is at a bank angle greater than or equal to 7 $\pm$ 1.0° in relation to the spin axis. Pin J2-F shall be connected to pin J2-A and the spin axis shall cease to precess in a simulated climb. The gyroscope assembly shall be returned to an attitude with a bank angle less than or equal to 4 $\pm$ 1.0° with respect to the spin axis, the connection between pins J2-A and J2-F shall be removed, and pin J1-H shall be connected to pin J2-A. The spin axis shall precess in a simulated climb and left bank as above but at a rate of not less than 15° per minute. Connecting pins J2-F and J2-K to J2-A shall cause precession to cease as specified above. During this test, the synchro outputs from the gyroscope shall give a direct indication of gyroscope position at all times.

4.8.5 Normal Erection Rate - The gyroscope shall be mounted on a low angle (1° total movement) roll, pitch, and yaw mechanism which operates at a speed of 5 to 7 oscillations per minute. The gyroscope shall be connected for normal operation and power applied. Starting from an approximate 10° climb position, the gyroscope shall be allowed to erect for 5° and the rate of erection shall be between 1.5° and 2.5° per minute with proper correction for earth's rotation. At high and low temperature extremes, the erection rate shall be between 1.0° and 3.0° per minute. The test shall be repeated for dive, left bank, and right bank positions, and the same rate limits shall apply. When the spin axis is displaced in one axis, it shall not deviate in the other axis by more than 1.0° while it is erecting.

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4.8.6 Manual Fast Erection Rate - The gyroscope shall be mounted on a level (within ± 1/2°) Scorsby table, as in 4.8.5. With the gyroscope connected for normal operation, power shall be applied. The spin axis shall be deflected 35° in either roll or pitch, and pin N (for the MD-1 replacement gyro) connected to pin C (manual fast erection), or pin J1-H (for the PB-60 replacement gyro) connected to pin J1-A (manual fast erection). The rate of erection from 35° to 5° of vertical shall not be less than 15° per minute. This test shall be performed clockwise and counterclockwise about each axis. When the spin axis is deflected in one axis, it shall not deviate in the other axis by more than 1.75° while it is erecting. The circuit between pins J and T (for the MD-1 replacement gyroscope assembly) or pins J1-M and J1-P (for the PB-60 replacement gyroscope assembly) shall be opened as long as the manual erection circuit is closed.

4.8.7 Erection Cutout

4.8.7.1 MD-1 (Replacement) Gyroscope Assembly - With the gyroscope in normal operation, the spin axis shall be precessed 10° in bank and 10° in pitch. Pin B shall be connected to pins K and L and there shall be no evidence of erection in either the pitch or roll axis. The current drawn from pin B to K and from pin B to L shall not exceed 250 ma. With the gyroscope in normal operation and erected to vertical, pins W and X shall be connected together. (Connect pin G to C and B to K to precess the spin axis in a simulated left bank attitude). Immediately cant the gyroscope assembly in a left bank attitude. The gyro shall cease to precess in a simulated left bank when the gyro assembly has been canted in a 7 ± 1.0° left bank attitude in relation to the spin axis. Cant the gyro assembly back to level. The spin axis shall begin to precess in a simulated left bank when the gyro assembly has been canted to within 4 ± 1.0° of the spin axis for 10 ± 5 seconds. Cant and maintain the gyro assembly in an approximately 10° left bank attitude with respect to the spin axis; the gyro shall cease to precess in a simulated left bank. After 5 ± 1 minutes from the time the spin axis exceeded 7 ± 1.0°, the spin axis shall begin to precess in a simulated left bank. Disconnect pin G to C and return the gyroscope assembly to level. The roll erection shall be allowed to operate until the spin axis is within 4 ± 1.0° of vertical in the roll axis. It shall be demonstrated that roll erection cutoff shall not be operative until the gyroscope is within 4 ± 1.0° for a period of 10 ± 5 seconds. Repeat the procedure canting the gyro assembly into a right bank instead of a left bank.

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4.8.7.2 PB-60 (Replacement) Gyroscope Assembly - With the gyroscope assembly in normal operation, the spin axis shall be precessed 5° in bank and pitch. Pin J2-F shall be connected to pin J2-A and the gyro shall erect in the roll but not the pitch axis. The current drawn from pin J2-A to pin J2-F shall not exceed 250 ma. The connection between J2-F and J2-A shall be removed; the spin axis shall erect in the pitch axis. With the gyroscope assembly in normal operation, connect pins J2-J to J2-A and J2-F to J2-A to precess the spin axis in a simulated left bank attitude. Immediately cant the gyro assembly in a left bank attitude; the gyro shall cease to precess in a simulated left bank when the gyro assembly has been canted in a 7 ± 1.0° left bank attitude in relation to the spin axis. Cant the gyro assembly back to level. The spin axis shall begin to precess in a simulated left bank when the gyro assembly has been canted to within 4 ± 1.0° of the spin axis for 10 ± 5 seconds. Cant and maintain the gyro assembly in an approximately 10° left bank attitude with respect to the spin axis; the gyro shall cease to precess in a simulated left bank. After 5 ± 1 minutes from the time the spin axis exceeded 7 ± 1 degrees, the spin axis shall begin to precess in a simulated left bank. Disconnect pin J2-J to J2-A and return the gyro assembly to level. The roll erection shall be allowed to operate until the spin axis is within 4 ± 1° of vertical in roll axis. It shall be demonstrated that roll erection cutoff shall not be operative until the gyroscope is within 4 ± 1° for a period of 10 ± 5 seconds. Repeat this procedure canting the gyro assembly into a right bank instead of a left bank.

4.8.8 Power Interruption - With the gyroscope assembly connected for 5 minutes for normal operation, power shall be removed (the circuit between J and T for the MD-1 replacement gyro assembly, of J1-M and J1-P for the PB-60 replacement gyro assembly, shall immediately open). The gyroscope shall be moved through angles of ± 60° in pitch and ± 75° in roll and returned to level. Power shall be applied 40 seconds after removal. The pitch and roll axis shall indicate level within 1°. The circuit between pins T and J (for the MD-1 replacement gyro assembly), or J1-P and J1-M (for the PB-60 replacement gyro assembly) shall immediately close, and there shall be no re-cycling of the initial erection.

4.8.9 Power Failure Warning

4.8.9.1 MD-1 (Replacement) Gyroscope Assembly - The gyroscope shall be connected for normal operation. During the initial erection period, there shall be no continuity between

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pin J and pin T. There shall be continuity between pin J and T immediately when the gyroscope switches from initial erection to normal running. If any phase of the 3 - phase power to the gyroscope is broken, the circuit between J and T shall immediately open.

4.8.9.2 PB-60 (Replacement) Gyroscope Assembly - The gyroscope assembly shall be connected for normal operation. The removal of input power between J1-A and J1-B shall immediately break the circuits between J1-P and J1-M. There shall be no continuity between J1-M and J1-P during the initial erection period.

4.8.10 Gimbal Freedom - With the gyroscope operating normally, displace it 360° in roll and ± 82° in pitch. There shall be no gimbal sticking or binding.

4.8.11 Drift - The vertical gyroscope shall be mounted on a roll, pitch, and yaw mechanism as in 4.8.5, with the forward end pointed to the geographical East. With the vertical gyroscope connected for normal operation, the top of the spin axis shall be precessed 20° in an Easterly direction and the erection circuits disconnected. The roll, pitch, and yaw mechanism shall be operated at a speed of 5 to 7 oscillations per minute with a total movement of ± 2 1/2°. With proper correction for the earth's rotation, the drift rate in either axis shall not exceed 15° per hour. Repeat the test with the forward end pointed to the South, North, and West with the spin axis 20° East. Repeat the above test with the spin axis at 0° tilt. During High and Low Temperature Operation Tests (4.8.23.1 and 4.8.23.2 respectively, and subparagraphs), the drift rate in either axis shall not exceed 30° per hour.

4.8.12 Power Consumption

4.8.12.1 MD-1 (Replacement) Gyroscope Assembly - With the gyroscope connected for normal operation, apply power and measure the current and voltage at start, 5 minutes after start with low erection on and with manual high erection on. The power consumption shall not exceed:

65 VA - Start
30 VA - Low Erection
55 VA - High Erection

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4.8.12.2 PB-60 (Replacement) Gyroscope Assembly - With the gyroscope connected for normal operation, apply power and measure the current and voltage at start, and at 10 minutes after start with low erection on. The power consumption shall not exceed:

70 VA - Start
40 VA - Low Erection

4.8.13 Synchro Requirements, Pitch and Roll Outputs

4.8.13.1 Synchro Output Voltages - With the gyroscope assembly connected to 115 V ± 0.25 V, (3-phase for the MD-1 replacement gyro assembly or single phase for the PB-60 autopilot replacement gyro assembly), 400 ± 1 Hz and operating normally, the output voltage at synchro null shall not exceed 30 mv when connected with up to five control transformers with normal impedance of 222 + j 470 ohms each (Zso), loaded with 10,000 ohms across the rotor. The maximum unloaded output voltage of the synchros shall be 11.8 ± 0.2 volts rms, and the phase shift shall not exceed 5.5°. When loaded with five loads the output voltage shall not be less than 10.4 volts rms and the phase shift shall not exceed 9°. At low temperature, unloaded, the output voltage shall be 10.8 ± 0.3 volts rms, phase shift shall not exceed 8.0°. With five loads the output voltage at low temperature shall not be less than 10.2 volts rms and the phase shift shall not exceed 11.5°. At high temperature, unloaded, the output voltage shall be 11.8 ± 0.3 volts rms, phase shift shall not exceed 8.0°. With five loads the output voltage at high temperature shall not be less than 10.2 volts rms and the phase shift shall not exceed 11.5°.

NOTE: Concerning synchro output voltages applicable to the PB-60 autopilot gyro assembly; the PB-60 assembly contains an internal load on both roll and pitch synchros. This constitutes one CT (control transformer) loading; therefore, voltage and phase shift tolerances may be compensated accordingly.

4.8.13.2 Static Accuracy - Static Accuracy shall be tested as follows:

a. Place the gyro on a table level to within 0.01°.

b. Rotate the gyro in the pitch axis in both climb and dive attitudes at 0°, ± 15°, ± 30°, ± 60°, and ± 75°. Record the pitch angle at each position as observed on an angle precision indicator (API). The accuracy specified in the schedule in paragraph 3.6.1.7.1 shall be met.

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c. After returning the gyro to level attitude, rotate the gyro in the roll axis in both right and left bank at 0°, ± 15°, ± 30°, ± 60°, and ± 180°. Record the roll angle at each position as observed on an API. The accuracy specified in the schedule in paragraph 3.6.1.7.1 shall be met.

4.8.13.3 Synchro Zeroing and Sensing

4.8.13.3.1 MD-1 (Replacement) Gyroscope Assembly - The snychros shall be zeroed and their lead pins shown in Table 1 connected for the following procedure.

a. Connect pins S and V to the ground side of a single-phase 115V, 400 Hz power source.

b. Connect pins M and U to the hot side of the power source.

c. Connect pins F and P to the common point of S - V. With the gyroscope gimbles at zero displacement about their axis, the voltage between pins D and E, and between pins R and H shall each be at a null.

d. With D and E connected, the voltage between this point and pin M shall be less than 115 V.

e. With pins R and H connected, the voltage between this junction and pin U shall be less than 115 V.

f. With pins D and E disconnected and the gyroscope displaced in right-bank about the roll axis with respect to the roll gimbal, the voltage between pin D and the common point of F - S shall increase before decreasing.

g. With pins R and H disconnected and the gyroscope displaced in climb about the pitch axis with respect to the pitch gimbal, the voltage between pin R and the common point of P-V shall increase before decreasing.

4.8.13.3.2 PB-60 (Replacement) Gyroscope Assembly - The synchros shall be zeroed and their lead pins on J1 shown in Table 1A connected for the following procedure:

a. Connect pins F and K to the common ground J2-B with the gyroscope gimbals at zero displacement about their axis, the voltage between pins D and E, and between pins J and H shall each be a null.

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b. With D and E connected, the voltage between this point and pin J2-A shall be greater than 115 V.

c. With pins J and H connected, the voltage between this junction and pin J2-A shall be greater than 115 V.

d. With pins D and E disconnected and the gyroscope displaced in right bank about the roll axis with respect to the roll gimbal, the voltage between pin E and the common point of F-K shall increase before decreasing.

e. With pins J and H disconnected and the gyroscope displaced in climb about the pitch axis with respect to the pitch gimbal, the voltage between pin H and the common point of F-K shall increase before decreasing.

4.8.13.4 Synchro Zeroing and Sensing - Alternate Test Method - (Applicable to MD-1 Replacement Gyroscope only). The pitch and roll gyroscope synchros zeroing and sensing may be tested using an indicator specified herein, or equivalent, when connected in accordance with Figure 1.

4.8.13.5 Autopilot Outputs - (Applicable to PB-60 Replacement Gyroscope Assembly).

4.8.13.5.1 Roll Output - The PB-60 (Replacement) Gyro Assembly shall be positioned in roll to obtain a null voltage between J1-e and J1-X with J1-X at ground. At null the assembly case shall be level within $\pm$ 1/2 degree. The null voltage shall not exceed 50 mv. Displacing the PB-60 gyro assembly $\pm$ 10 degrees $\pm$ 0.1° in bank from reference null position shall produce a signal between J1-e and J1-X of 2.0 $\pm$ 0.2 volts. A right bank displacement shall produce an in phase signal within $\pm$ 5.5° when loaded with 10K ohms.

4.8.13.5.2 Pitch Output - The PB-60 (Replacement) Gyro Assembly shall be positioned in pitch to obtain a null voltage between J1-G and J1-V with J1-V at ground. At null the assembly case shall be level within $\pm$ 1/2 degree. The null voltage shall not exceed 50 mv. Displacing the PB-60 assembly $\pm$ 10 degrees $\pm$ 0.1° from the reference null position shall produce a signal between J1-G and J1-V of 2.0 $\pm$ 0.2 volts. A climb displacement shall produce an in phase signal within $\pm$ 5.5° when loaded with 10K ohms.

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4.8.13.5.3 Versine Output - The PB-60 (Replacement) Gyro Assembly shall be positioned in pitch and roll to the reference null positions. The null voltage measured on a VTVM between J1-S and J1-T with L1-T at ground shall not exceed 150 mv. If null exceeds 150 mv, position quadrant selector of phase sensitive voltmeter (psvm) to 90 or 270 degrees and calibration dial at zero. Quadrature voltage shall not exceed 150mv. The in-phase voltage component reading on the psvm should be not greater than 480 mv at 0°. Displacing the assembly case to ± 20 degrees in bank shall produce an in-phase signal of 360 ± 72 mv and phase shift of ± 5.5 degrees maximum. Tilt the assembly case to ± 30 degrees in bank. The voltage reading on the psvm, for either position, should be 800 ± 160 mv and phase ± 5.5 degrees maximum.

4.8.14 Insulation Resistance

4.8.14.1 MD-1 (Replacement) Gyroscope Assembly - With a potential of 500 VDC applied for 5 seconds between receptacle pins, and between receptacle pins and case, as listed below, the resistance shall be greater than 50 megohms. This test shall be conducted only one time on any one gyroscope.

Pin	Case	U	T	M	J	H	D
A	X	X	X	X	X	X	X
D	X	X	X	X	X	X	
H	X	X	X	X	X		
J	X	X	X	X			
M	X	X	X				
T	X	X					
U	X						

NOTE: This test shall not be applied to circuits connected to electrolytic switches.

4.8.14.2 PB-60 (Replacement) Gyroscope Assembly - With a potential of 500 VDC applied for 5 seconds between receptacle pins and between receptacle pins and case, the resistance shall be greater than 50 megohms. This test shall be conducted only one time on any one PB-60 Autopilot replacement gyroscope assembly. This test shall not be performed on pins which are connected to the erection circuits.

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4.8.15 Leakage Test - The gyroscope assembly shall be tested for leaks with a helium leak detector. The leak rate shall not be sufficient to cause more than 10 percent loss or contamination of the filling medium after 1000 hours at a pressure differential of one atmosphere.

4.8.16 Power Variation - The gyroscope assembly shall be tested to the requirements of Paragraph 4.8.2, 4.8.3, and 4.8.9 of this specification at voltage and frequency settings of 103V at 380 Hz and 127V at 420 Hz. The gyroscope assembly shall be tested to the requirements of Paragraph 4.8.10 and 4.8.11 at 103V, 380 Hz.

4.8.17 Dielectric Strength

4.8.17.1 MD-1 (Replacement) Gyroscope Assembly - A potential of 500 volts (root mean square) alternating current at 60 Hz shall be applied between receptacle pins, and between receptacle pins and the case ground as listed below for a period of 5 seconds. There shall be no breakdown of insulation or any other permanent damage to the instrument as a result of this test. Insulation breakdown shall be defined as an alternating current leakage impedance of less than 200,000 ohms, or a leakage current of 2.5 milliamperes.

Pin	Case	U	T	M	J	H	D
A	X	X	X	X	X	X	X
D	X	X	X	X	X	X	
H	X	X	X	X	X		
J	X	X	X	X			
M	X	X	X				
T	X	X					
U	X						

NOTE: This test shall not be applied to circuits connected to electrolytic switches.

4.8.17.2 PB-60 (Replacement) Gyroscope Assembly - A potential of 500 volts (root mean square) alternating current at 60 Hz shall be applied between receptacle pins, and between receptacle pins and the case ground for a period of 5 seconds. There shall be no breakdown of insulation or any other permanent damage to the instrument as a result of this test. Insulation breakdown shall be defined as an alternating current leakage impedance of less than 200,000 ohms, or a leakage current of 2.5 milliamperes. This test shall not be performed on pins which are connected to the erection circuits.

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4.8.18 Loop Error - The gyroscope shall be mounted on a fixture permitting rotation about the pitch axis. The unit shall be rotated one complete revolution (loop) at a rate of 10° ± 1° per second, and returned to the normal position. The error in either pitch or roll shall not exceed 3°.

4.8.19 Roll Test - The gyroscope shall be mounted on a fixture permitting rotation about the roll (longitudinal) axis. The gyroscope shall be rotated 5 complete revolutions (roll) at a rate of 400° ± 10° per second. The error induced in either pitch or bank shall not exceed ± 1°.

4.8.20 Vibration Error - The vibration error test shall consist of a frequency survey with vibration applied to longitudinal, lateral, and vertical axis of the gyroscope. The gyroscope shall be mounted with the base plate and shall be subjected to vibration with a constant applied double amplitude of 0.060 inch through the frequency range of 10 to 55 to 10 Hz within one minute in each of three mutually perpendicular axis. This test may be conducted with vibration applied in a circular motion for one complete frequency range.

4.8.20.1 Signal Continuity - There shall be no transients or discontinuity in the pitch and roll synchro outputs during vibration.

4.8.20.1.1 Synchro Transmitter Characteristics - After vibration, the electrical characteristics of the transmitting synchro shall be measured with suitable test equipment. The characteristics shall be within the tolerances specified in 4.8.13.

4.8.20.2 Vibration - After the one-minute vibration, the pitch and roll axis shall not be more than 3/4° from their original position before vibration.

4.8.21 Gimbal Clearance - After having passed the test specified in 4.8.10, the gyroscope shall be subjected to a differential pressure of one atmosphere greater than the pressure inside the gyro case. The pressure differential shall be maintained for 15 minutes. At the conclusion of this test, the gyroscope shall again be subjected to the test specified in 4.8.10.

4.8.22 Electromagnetic Interference - The gyroscope shall be tested in accordance with MIL-STD-462, Notice 2, Test Methods

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CE03, CE04, CS01, CS02, CS06, RE02, RS02, and RS03 to demonstrate compliance with the requirements of paragraph 3.4.5 of this equipment specification.

4.8.23 Environmental Tests - The gyroscope assembly shall be subjected to the following environmental tests conducted in accordance with the applicable methods of MIL-STD-810 and as specified herein.

4.8.23.1 High Temperature

4.8.23.1.1 High Temperature Operation - The gyroscope shall be properly connected and placed in a chamber at a temperature of +85° ± 2°C for 4 hours. After the soak period, power to the gyro shall be terminated and the gyro rotor permitted to run down to zero rpm. The high temperature shall be maintained during rotor run down and during subsequent tests. After rotor run down, the gyroscope shall be subjected to and shall meet the initial erection, drift, and synchro transmitter characteristics as specified in paragraphs 4.8.2, 4.8.11, and 4.8.13.1, respectively. The limits and tolerances applicable to the synchro transmitter output and phase shift shall be as specified in 4.8.13.1 for high temperature operation, for the unloaded and loaded conditions.

4.8.23.1.2 High Temperature Exposure - The gyroscope shall be properly connected and subjected to a high temperature exposure test in accordance with Method 501, except that the exposure period shall be 20 hours and no power applied. At the end of the 20 hour period and while at the high temperature, the gyroscope shall be subjected to the tests specified in 4.8.2, 4.8.3, 4.8.9, 4.8.10, and 4.8.11. The temperature shall be raised to 100° ± 2°C within 15 minutes, maintained for 30 minutes, then reduced to 71° ± 2°C within 15 minutes and maintained for 2 hours and 50 minutes. The gyroscope shall then be tested in accordance with 4.8.10.

4.8.23.2 Low Temperature

4.8.23.2.1 Low Temperature Operation - The gyroscope shall be placed in a chamber at a temperature of -54° ± 2°C for a period of four hours. At the end of this period and while the gyroscope is at the reduced temperature, the tests for initial erection, final erection, erection rate, drift synchro transmitter characteristics, gimbal freedom, and power warning circuit shall be repeated (reference paragraphs 4.8.2, 4.8.3, 4.8.5, 4.8.11, 4.8.13.1, 4.8.10, and 4.8.9 respectively). The

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gyroscope shall meet the requirements for these tests with the following exceptions:

4.8.23.2.1.1 Before conducting the initial erection tests, the spin axis shall be displaced at any random pitch angle within ± 82° and any random roll angle within ± 80° from the vertical at the time power is applied. The initial erection time shall not exceed 1 1/2 minutes after power is applied.

4.8.23.2.1.2 Power shall be applied for 10 minutes before conducting the final erection test.

4.8.23.2.1.3 The normal erection rate limits shall be between 1.0° and 3.5° per minute with a 1° maximum deviation on the undeflected axis.

4.8.23.2.1.4 The no load output shall be 11.8 ± 0.3 volts rms and the no load phase shift shall not exceed 8°. The line-to-line output voltage when loaded with five loads as specified in 3.5.1.7, shall not be less than 10.2 volts rms. The phase shift with five loads shall not exceed 11.5°.

4.8.23.2.2 Low Temperature - High Altitude - The gyroscope shall be subjected to a low temperature-high altitude test in accordance with Methods 500 and 502. The low temperature (including low operating temperature) for both methods shall be -62°C. Method 500 testing shall be performed at the end of the 48 hour low temperature test period. The pressure altitude shall be equivalent to 100,000 ± 500 feet. With power applied and with the temperature and pressure as specified above, the initial erection and gimbal freedom tests shall be performed. The time required for initial erection shall not exceed 5 minutes. There shall be no binding or sticking of the gimbals at any indication. After the gyroscope has returned to room temperature and pressure, it shall be subjected to and shall meet the requirements of all of the individual tests.

4.8.23.3 Humidity - The humidity test shall be conducted in accordance with Method 507, Procedure I. The gyroscope shall then meet the leak rate test. There shall be no evidence of corrosion or rust and the gyroscope shall be subjected to and meet the performance requirements of the individual tests.

4.8.23.4 Fungus - The fungus test shall be conducted in accordance with Method 508. The gyroscope shall then meet the leak rate test. There shall be no deterioration nor shall any

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part of the gyroscope support fungus growth, and the gyroscope shall be subjected to and meet the performance requirements of the individual tests.

4.8.23.5 Salt Fog - The salt fog test shall be conducted in accordance with Method 509. The gyroscope shall be subjected to and meet the performance requirements of the individual tests.

4.8.23.6 Vibration Endurance - The gyroscope and base plate assembly shall be subjected to vibration testing in accordance with Method 514.2, Equipment Category (a), Test Procedure I, Table 514.2 - II, Figure 514.2-2, Curve B. At the completion of this test, the gyroscope shall be subjected to and meet the performance requirements of the individual tests.

4.8.23.7 Acceleration - The gyroscope shall be subjected to 20g acceleration for over one minute along each of its three axes while non-operating. The gyroscope shall then be subjected to and meet the requirements of the individual tests.

4.8.23.8 Shock - The shock tests shall be conducted in accordance with Method 516.1, Procedures I and III, Figure 516.1.1, parameters (a) and (c). The gyroscope shall be subjected to Procedure I and meet the requirements of the individual tests. A dummy gyroscope assembly shall be subjected to and meet the requirements of Procedure III.

4.8.23.9 Temperature Shock - The gyroscope shall be subjected to a temperature shock test in accordance with Method 503. After the gyroscope has returned to room temperature, the connector and seals shall be examined. There shall be no evidence of cracked terminals or leaks at the seals. The gyroscope shall then be subjected to the test specified in 4.8.11.

4.8.24 Compatibility (See 6.7) - Prototype installation and flight compatibility tests shall consist of the following: Initial erection, final erection, erection rate, manual fast erection, power interruption, and power warning circuit. As many of these tests as possible shall be conducted on the ground prior to aircraft takeoff. The rest of the tests shall be conducted in the air during straight and level flight and during coordinated turns.

4.8.25 Reliability Qualification Test

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4.8.25.1 Reliability qualification shall consist of successfully meeting the requirements for an accept decision in accordance with Test Plan XVIII, MIL-STD-781, under the conditions specified in 4.8.25.2. Specified MTBF shall be θo = 4200 hours. This test shall be performed using a quantity of fifteen (15) gyroscopes, five of which shall be the Type PB-60 Autopilot (Replacement) Gyroscope Assembly configuration, and ten of which shall be of the MD-1 (Replacement) Gyroscope Assembly configuration.

4.8.25.2 During this test, the gyroscope shall be properly connected and operated on a scorsby-type mechanism having a 15° movement at 5 to 7 oscillations per minute. Test level F of MIL-STD-781 shall be used, except that the vibration shall be 1 G and the temperature extremes shall be +35°C and -10°C. The gyroscope shall be operated for a period of 20 hours followed by an off time of 4 hours. This time cycle shall be repeated using each gyroscope until successful completion of Test Plan XVIII, MIL-STD-781. During this test, the performance of each gyroscope shall be verified by the tests specified in 4.8.2 (Initial Erection), 4.8.3 (Final Erection), 4.8.4 (External Torquing), and 4.8.11 (Drift) at lease once within each 200 hours of individual gyroscope operation. Only equipment "on" time may be used in MTBF determinations. In the event of a failure, the equipment may be repaired and returned to test at the option of the contractor in order to insure timely completion of the test.

4.8.26 Dynamic Brake Test - The gyroscope shall be subjected to a dynamic brake test as follows:

a. A source of 24 volts DC is applied between pins Q(+) and Z(-) of the gyroscopes connector. The gyroscope shall be brought to normal operating speed by application of AC power in accordance with 4.7.4 and maintained for 5 minutes. At the end of this time, AC power shall be interrupted. The gyro-wheel spin-motor shall slow rapidly as a result of the dynamic braking action. Evidence of braking action can be an audible indication of wheel speed slowing, and/or a drain of 0.4 to 2 amperes from the DC supply connected to pins Q(+) and Z(-) while the braking action is taking place. After 60 seconds (from the time of AC power interruption), dynamic brake reset shall be indicated by a drop in this loading of the DC source to a value less than 0.1 ma. The AC power shall then be reapplied and the gyro shall demonstrate successful fulfillment of Initial Erection and Final Erection requirements in accordance with 4.8.2 and 4.8.3.

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b. The test from a. should be repeated, except that braking action shall be interrupted after 40 seconds (approximately 20 seconds before automatic resetting) by reapplication of AC power. The tests of 4.8.2 and 4.8.3 shall again be performed.

4.9 Packaging Inspection.

4.9.1 Quality Conformance. The inspection of the preservation-packaging and interior package marking shall be in accordance with group A and B quality conformance inspection requirements, section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.9.2 First Article. When the unit container is capable of serving as the shipping container First Article Inspection, and rough handling tests as outlined in section 4 of MIL-P-116, shall be accomplished followed by a functional test of the unit to insure freedom from operational malfunction.

5. PACKAGING

5.1 Preservation-Packaging. Preservation-packaging shall be level A, B, or C as specified by procuring activity (see 6.4).

5.1.1 Level A.

5.1.1.1 Cleaning. Unit shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Immediately after cleaning, the unit shall be dried following any one or combination of the drying procedures of MIL-P-116. The drying procedures employed shall not be injurious to the unit.

5.1.1.3 Preservation Application. Not required.

5.1.1.4 Unit Packaging. Unless otherwise specified by the procuring activity, each unit shall be packaged in quantity unit packs of one each in accordance with method 1C of MIL-P-116. Overbox each completed pack in a PPP-B-636 Fiberboard, weather-resistant class box. Cushioning material between bag and container shall be of a type, density and thickness that will protect the unit from impact shock damage when tested as specified in paragraph 4.9.2 of this specification.

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5.1.2 Level B. Each unit shall be packaged as indicated for Level A except that a specification PPP-B-636 class domestic box shall be used as the unit container.

5.1.3 Level C. Units shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.4).

5.2.1 Level A. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636 class weather-resistant. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, specification PPP-B-601 plywood box, shall be used for the shipping container. When specified by the procuring activity, the container shall conform to MIL-C-4150.

5.2.2 Level B. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636, class domestic. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, the shipping containers shall conform to specification PPP-B-636 class weather resistant.

5.2.3 Level C. Packaged units shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use - The gyroscope covered by this specification is intended to describe performance requirements for a roll and pitch gyroscope.

6.2 Test Values - Normal and limiting values of performance data shall be determined at an input voltage of 115 $\pm$ 0.5 VAC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

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6.3 Gyroscope Definition - When referring to "The Gyroscope" it shall mean a complete system, a gyroscope assembly, electronics assembly and mounting base assembly. Throughout this specification, all requirements apply to both MD-1 (Replacement) Gyroscope Assembly and PB-60 (Replacement) Gyroscope Assembly, unless specifically addressed to one or the other.

6.4 Ordering Data - Purchasers should exercise any desired options offered herein, and procuring documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

c. When Sampling Plan A is not to be conducted (see 4.4.2.2.1).

6.5 Qualified Products Lists - With respect to products requiring qualification testing, awards will be made only for such products as have, prior to time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products Lists whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that the propose to offer to the Federal Government tested for approval in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products Lists is the Engineering Division, ATTN: MMEDO, Oklahoma City ALC, Tinker AFB OK 73145. Information pertaining to qualification of products may be obtained from that activity.

6.6 Installation - Installation of the MD-1 (replacement) gyroscope is covered by Figure 6. Installation of the PB-60 Autopilot (replacement) gyroscope assembly is governed by Figure 9.

6.7 Compatibility - The prototype installation and compatibility tests will be performed by the Air Force with technical support and assistance of the contractor. Changes or modifications resulting from these tests will be accomplished by the contractor on all production gyros at no extra cost to the Government. Any gyro delivered prior to completion of satisfactory flight tests will be reworked and updated at no extra cost to the Government.

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6.8 Installation with DC Brake Circuit - Installation of the MD-1 (replacement) gyroscope and/or the PB-60 Autopilot (replacement) gyroscope assemblies on aircraft utilizing the DC brake circuit feature of the gyroscope shall have the 24 volts from the aircraft battery (or auxiliary DC supply) applied to the gyroscope connector on pins Q(+) and Z(-) through a circuit accessibility device, such as the aircraft landing gear switch. This is to prevent accidental brake application during flight by making the brake voltage accessible to the gyro only while the aircraft is on the ground. Only the positive (+) potential of the brake voltage need be switched in this manner.

Custodian:
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Air Force - 99

Project Number:
6615-F040

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TABLE 1

Pin	Function
A	Power Ground)
B	115V, 400Hz) See Note 1
C	115V, 400Hz)
D(X)	
E(Y)	Roll Transmitter Synchro Stator
F(Z)	
R(X)	
H(Y)	Pitch Transmitter Synchro Stator
P(Z)	
G	External Torquing) See Note 2
J	Power Warning Flag
T	Power Excitation for Alarm Flag
K	Pitch Erection Cutout
L	Roll Erection Cutout
M(H)	Roll Erection Excitation
S(C)	
U(H)	Pitch Synchro Excitation
V(C)	
Y	Single Phase Power Warning Enable (Connect to Pin A)
N	Manual Fast Erection
W	Bank Angle Roll Erection Cutout Enable
X	Bank Angle Roll Erection Cutout Enable
Q	Dynamic Brake, $\pm$28V DC
Z	Dynamic Brake, 28V DC Return

Note 1: The input power to the gyro may be 115V, 400Hz, single phase per Figure 2; or 115V, 400Hz, 3 phase delta, phase rotation on Pins A, B, and C per Figure 1. Pin A shall be power ground. Phase rotation shall be Pin C to A to B.

Note 2: When Pin G is utilized, the short between Pins W and X, if existing, should be removed.

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TABLE 1A

J1 Pin	Function
A	Spare
B	Spare
C	Spare
D(Y)	Roll Transmitter Synchro
E(X)	Roll Transmitter Synchro
F(Z)	Roll Transmitter Synchro
G(H)	Pitch A/P 200 mv/deg @0°
H(X)	Pitch Transmitter Synchro
J(Y)	Pitch Transmitter Synchro
K(Z)	Pitch Transmitter Synchro
L	Spare
M	A/P Interlock
N	Spare
P	A/P Interlock
R	Dynamic Brake, +28V DC
S(H)	Versine 6000 (1 - Cos θ) mv/deg @ 0°
T(C)	Versine
U	Spare
V(C)	Pitch A/P
W	Spare
X(C)	Roll A/P
Y	Spare
Z	Dynamic Brake, 28V DC Return
a	Spare
b	Spare
c	
d	
e(H)	Roll A/P 200 mv/deg @ 0°
f	
g	
h	Manual Fast Erection Input (115V @ 0°)
j	Spare

J2 Pin	Function
A(H)	Power In, 115V, 400Hz (Ref)
B(C)	Power Ground
C	Power Ground
D	
E	

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TABLE 1A (Continued)

J2 Pin (Continued)	Function
F	Pitch Erection Cut-Out Input, (115V @ 0°)
G	
H	
J	External Torquing Input (115V @0°)
K	External Roll Erection Cut-Out Input (115V @0°)

J3 Pin	Function
A	Power Ground
B	115V, 400Hz
C	
D(X)	Roll Transmitter Synchro
E(Y)	Roll Transmitter Synchro
F(Z)	Roll Transmitter Synchro
R(X)	Pitch Transmitter Synchro
H(Y)	Pitch Transmitter Synchro
P(Z)	Pitch Transmitter Synchro
G	External Torquing
J	Power Warning Flag
T	Power Warning Flag
K	Pitch Erection Cutout
L	External Roll Erection Cutout
M(H)	Roll Synchro Excitation
S(C)	Roll Synchro Excitation
U(H)	Pitch Synchro Excitation
V(C)	Pitch Synchro Excitation
N	Bank Angle Roll Erection Cutout Enable
Y	Power Failure Warning, Single Phase, Enable
Q	Dynamic Brake, +28V DC
Z	Dynamic Brake, 28V DC Return

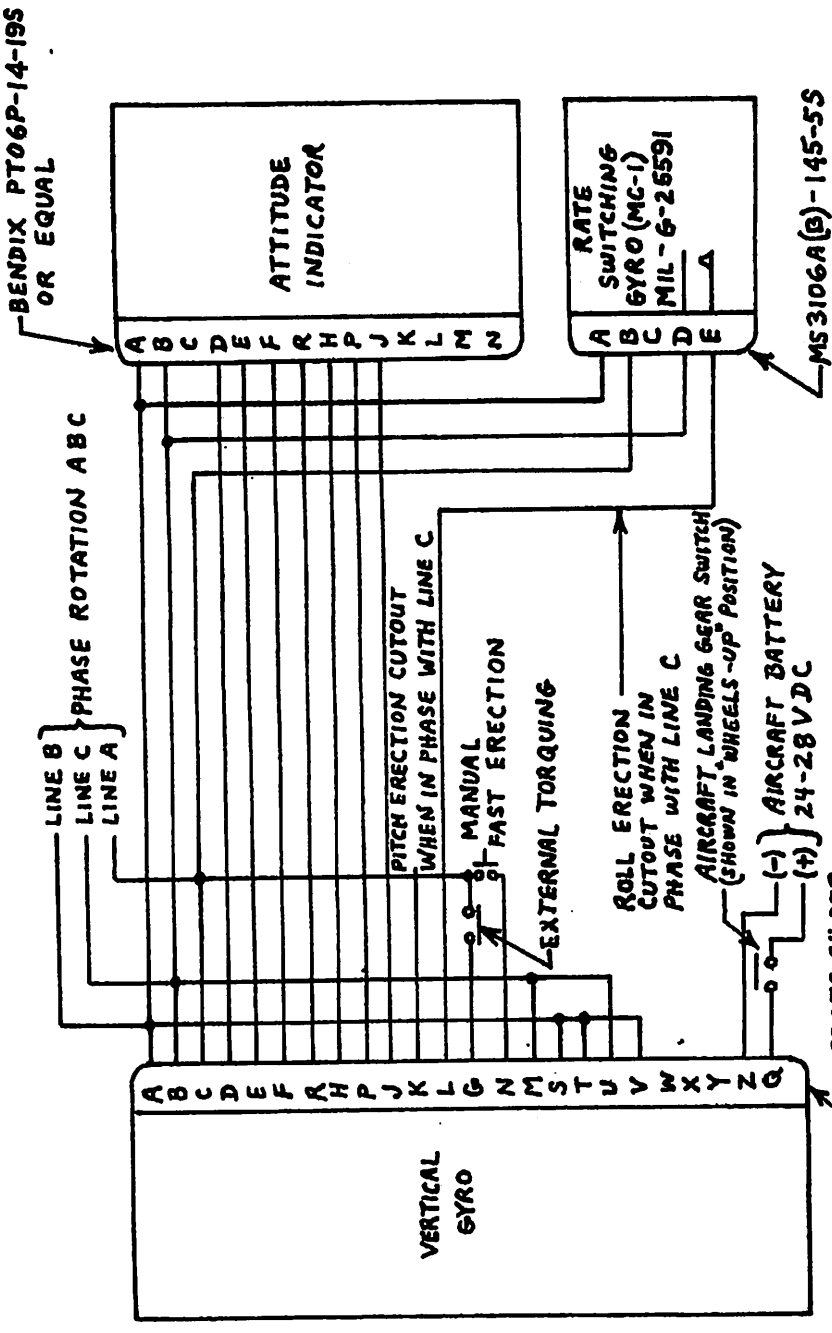
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TABLE 1B

COMMON PINS ON PB-60 BASE-PLATE ASSEMBLY

J3-A	J2-B
J3-B	J2-A
J3-Y	J2-B
J3-D	J1-E
J3-E	J1-D
J3-F	J1-F
J3-R	J1-H
J3-H	J1-J
J3-P	J1-K
J3-G	J2-J
J3-J	J1-M
J3-T	J1-P
J3-K	J2-F
J3-L	J2-K
J3-M	J2-B
J3-S	J2-A
J3-U	J2-B
J3-V	J2-A
J3-N	J1-h
J3	J1-R
J3-	J1-Z

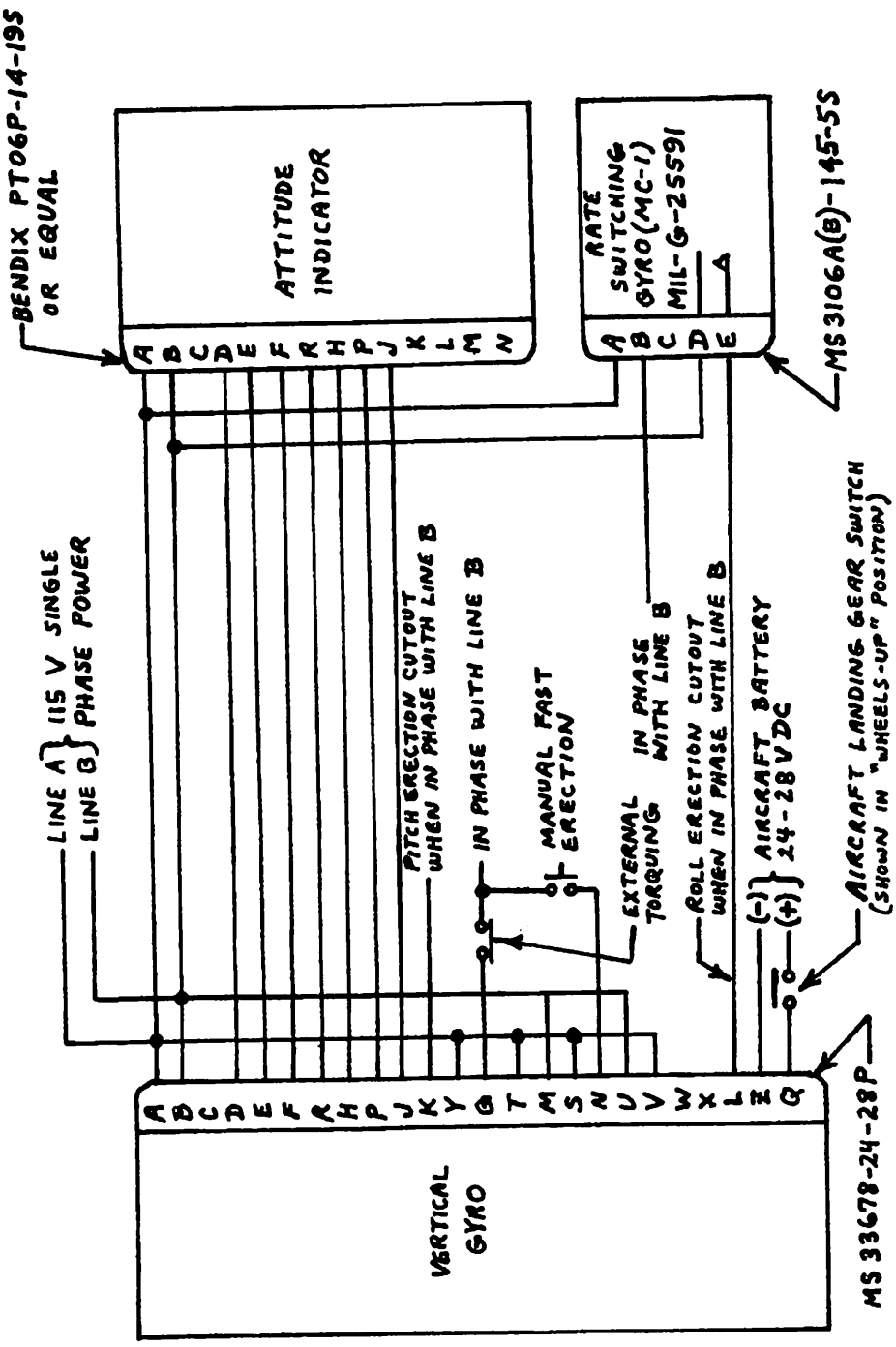
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SCHEMATIC, 3 PHASE

FIG 1

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SCHEMATIC, SINGLE PHASE

FIG 2

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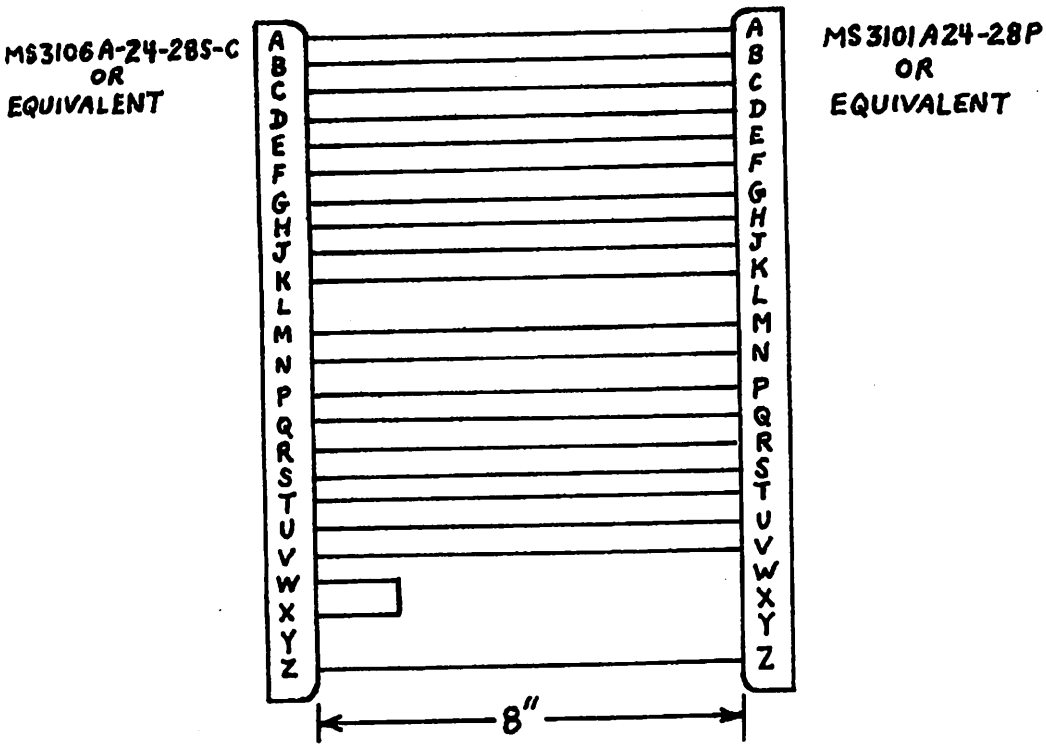


FIG 3 ADAPTOR CABLE SCHEMATIC

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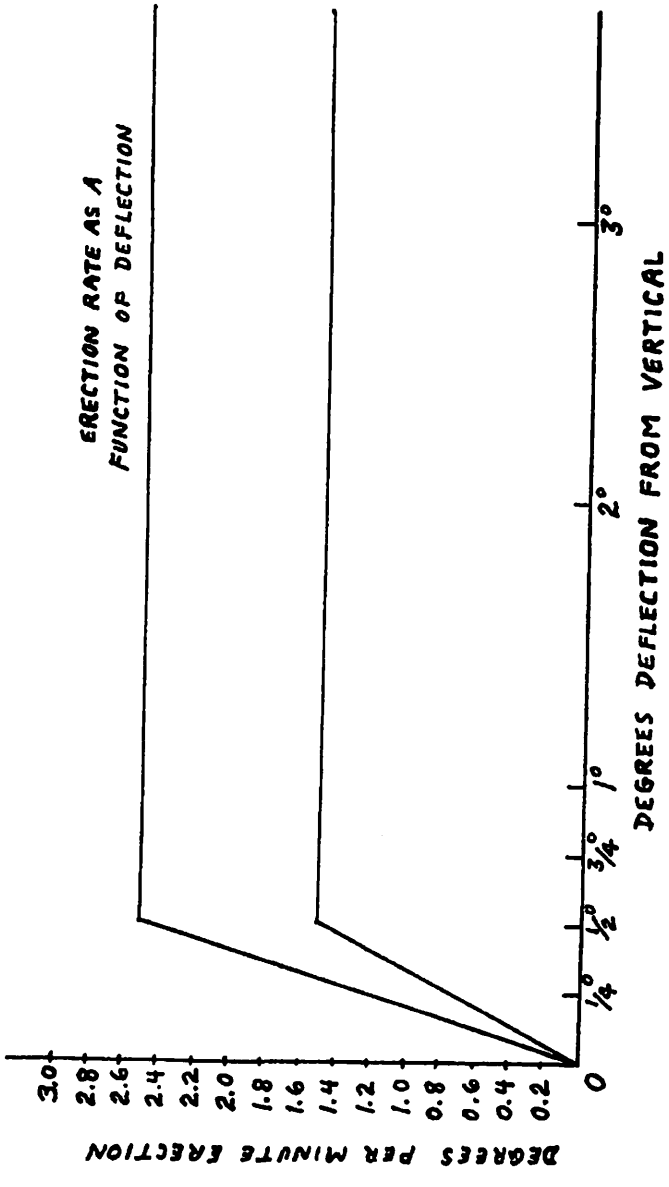


FIGURE 4. LIMITS

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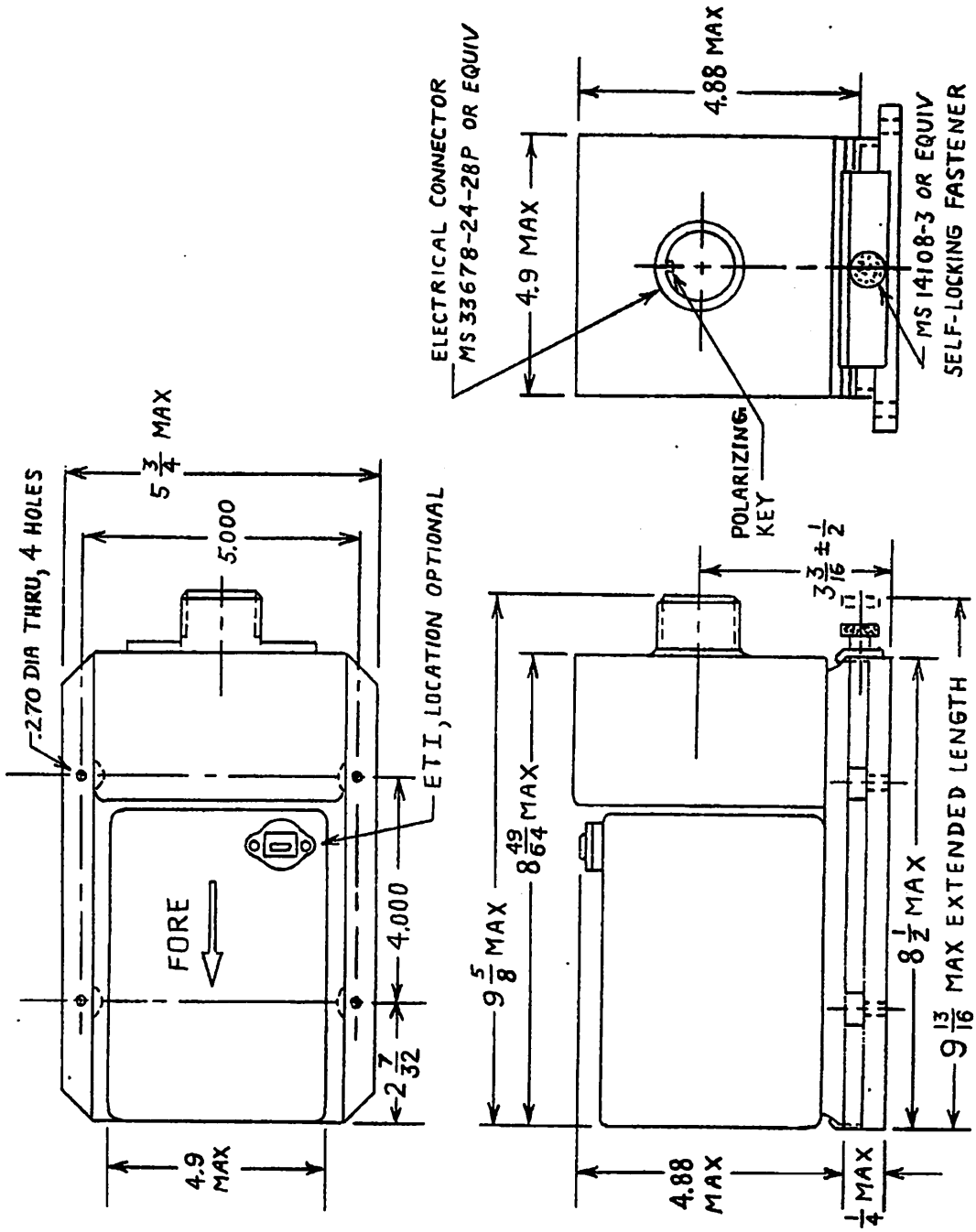
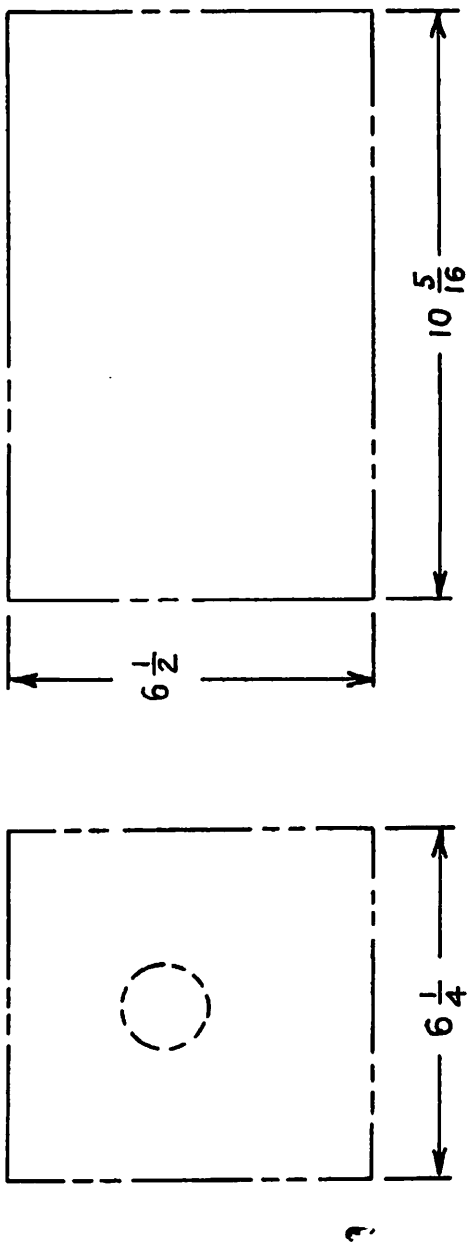


FIG 5 GYRO & BASE (MD-1 REPLACEMENT)

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INSTALLATION CLEARANCE OUTLINE (MD-1 REPLACEMENT)

FIG 6

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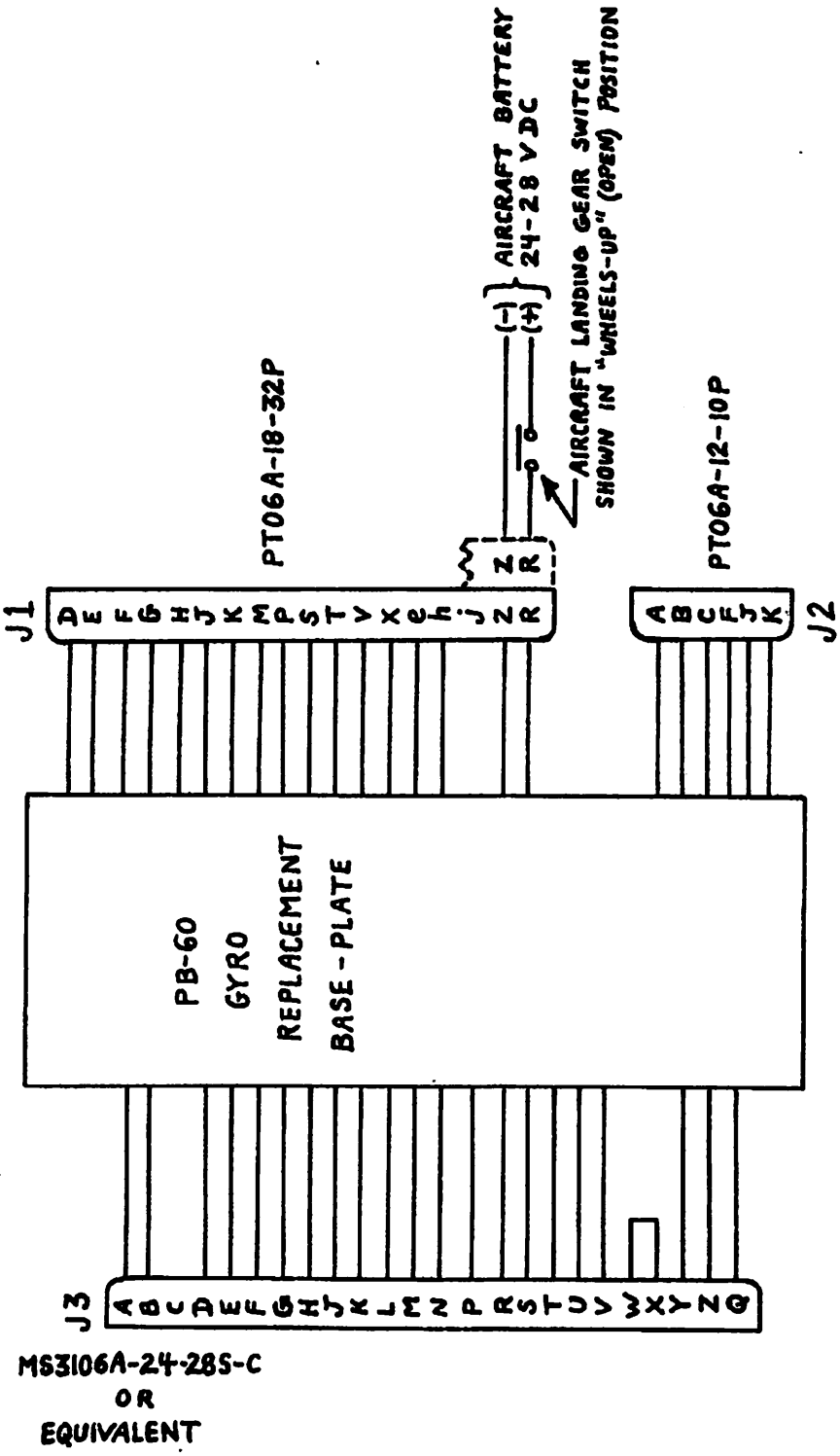
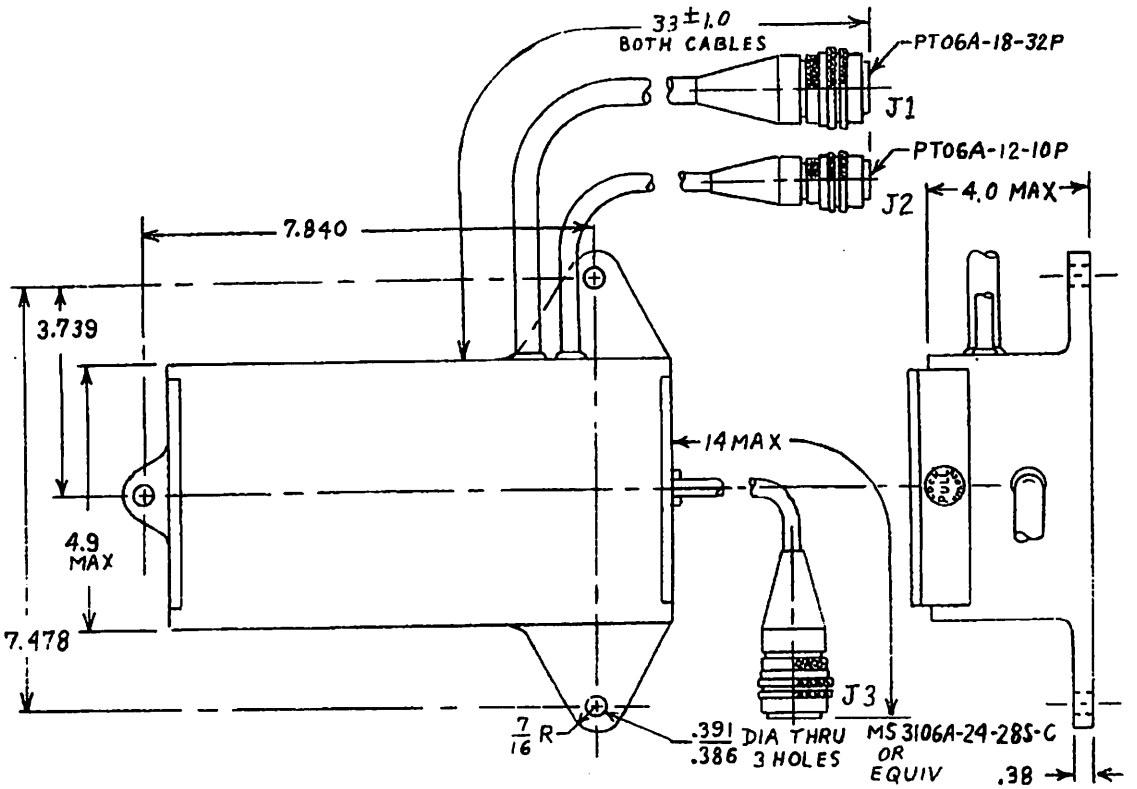
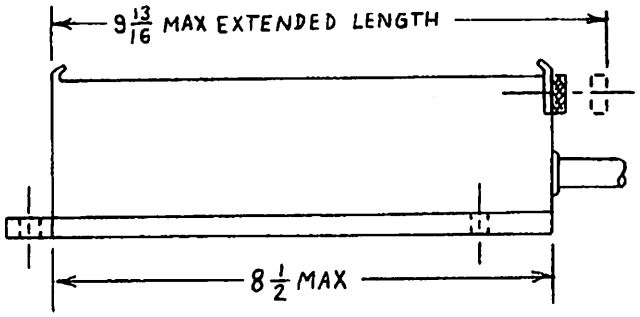


FIG. 7 PB-60 GYRO REPLACEMENT BASE-PLATE SCHEMATIC

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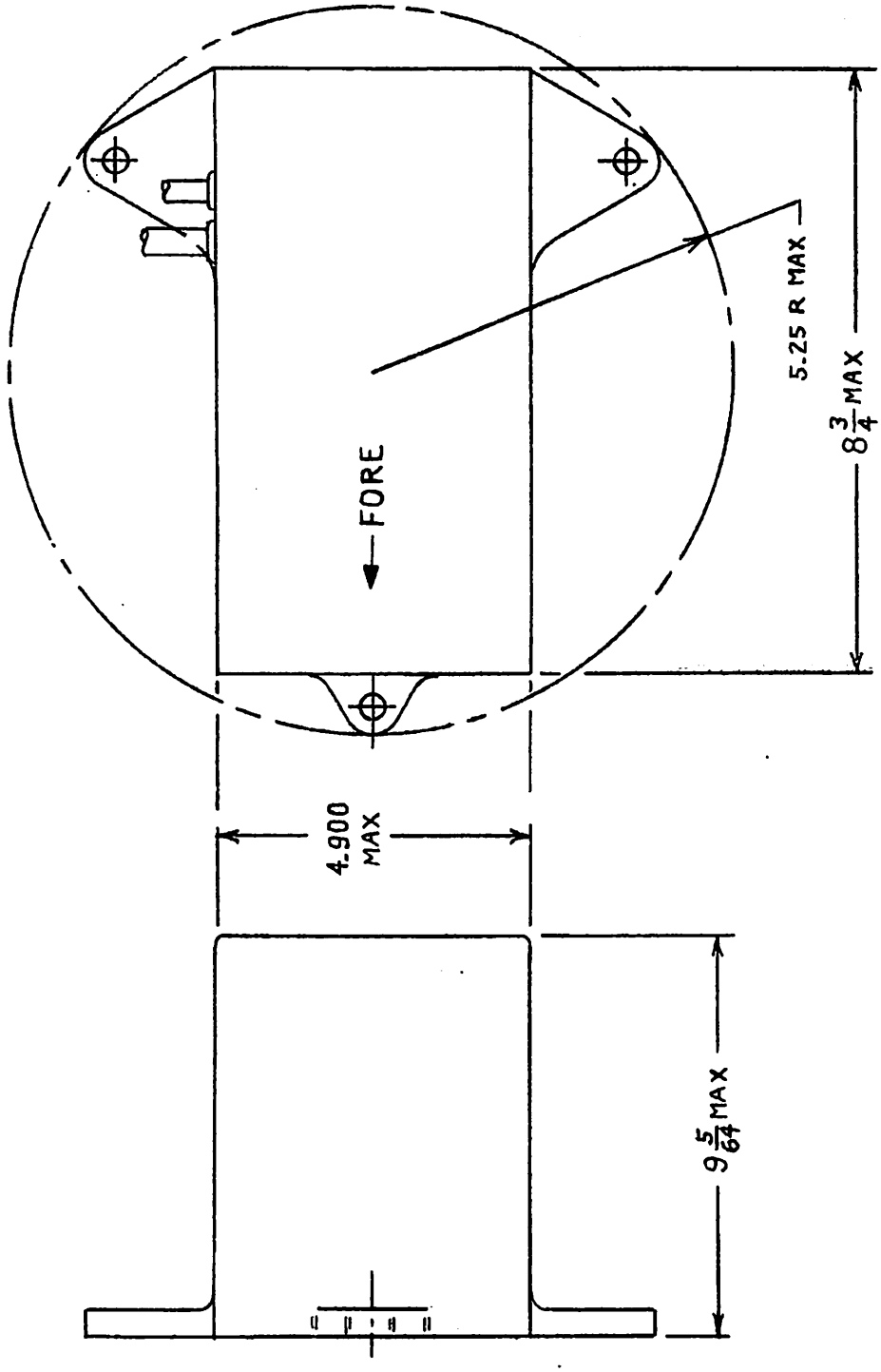




PB-60 (REPLACEMENT) BASE-PLATE OUTLINE DWG

FIG. 8

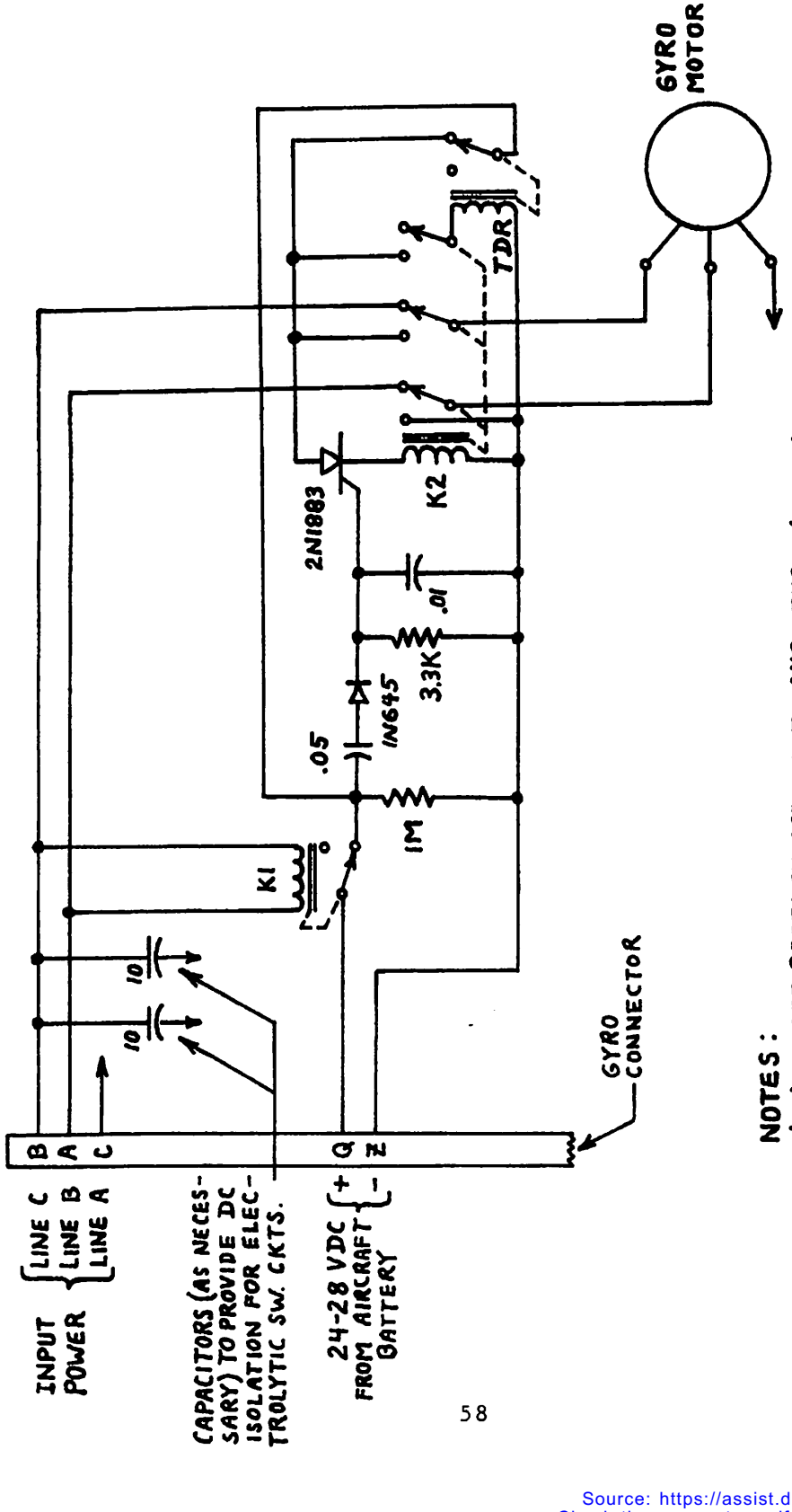
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INSTALLATION CLEARANCE OUTLINE (PB-60)

FIG. 9

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NOTES:
1. AIRCRAFT BATTERY APPLIED TO GYRO THRU LANDING GEAR SWITCH (TO PREVEND DC BRAKING WHILE AIRBORN).
2. TDR DELAYS ON PULL-IN (APPROX 60 SEC)
3. K1, K2 SHOWN IN DE-ENERGIZED STATE.
4. SCHEMATIC ABSTRACTED FROM U.S. PAT. 3903464 TO WHICH THE GOVERNMENT HAS ROYALTY FREE LICENSE.

FIG 10. EXAMPLE CIRCUIT FOR AUTOMATIC, RESETTING, DYNAMIC BRAKE

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